

ACADEMIC HEALTH CENTERS AND SOCIAL DETERMINANTS OF HEALTH:
COMMUNITY HEALTH IMPROVEMENT

Marc Caruana

A dissertation submitted to the faculty at the University of North Carolina at Chapel Hill in partial fulfillment of the requirements for the degree of Doctor of Public Health in the Department of Health Policy and Management of the Gillings School of Global Public Health.

Chapel Hill
2023

Approved by:

Morris Weinberger

Sandra Greene

David Flynn

Christopher Shea

Alexandra Zuber

© 2023
Marc Caruana
ALL RIGHTS RESERVED

ABSTRACT

Marc Caruana: Academic Health Centers and Social Determinants of Health:
Community Health Improvement
(Under the direction of Morris Weinberger)

Problem: Academic Health Centers (AHCs) have traditionally focused on their three core missions of patient care, research, and education. The current changing health care environment requires AHCs to continue to offer specialized tertiary care services to individual patients while also broadening their focus to improve the health of their communities.

Study Objective: The objective of this study is to identify the opportunities and challenges for AHCs in North Carolina that are addressing Social Determinants of Health (SDOH) to improve the health of patients and their communities. The five AHCs in North Carolina during the study period were the University of North Carolina Health, Duke University Health System, Vidant Health System, Wake Forest Baptist Health, and Atrium Health.

Data Collection: The study used a mixed methods design. Financial data from system wide community benefits reports and the Form 990 Schedule H of the primary medical centers (when available) from the five AHCs in North Carolina were analyzed. Additionally, four key informant interviews were completed at each of the five AHCs in North Carolina.

Plan for Change: A proposed framework for a plan for change incorporates the eight steps in the Kotter process for leading change. Leaders and an army of volunteers will be key to

develop partnerships with local organizations addressing SDOH. A new strategic vision must incorporate social accountability into their core missions to remove barriers and institute and sustain change. The long-term goal is to establish and sustain partnerships with active programs addressing SDOH to improve the health of their community through community building activities.

TABLE OF CONTENTS

LIST OF FIGURES	ix
LIST OF TABLES	x
LIST OF ABBREVIATIONS.....	xi
CHAPTER 1: INTRODUCTION AND BACKGROUND	1
Background	1
AHC Missions.....	5
Value-Based Models	6
Community Characteristics.....	9
Conceptual Framework	10
Aims and Research Question	11
CHAPTER 2: LITERATURE REVIEW	13
Information Sources.....	13
Reference Selection	14
Summary of Findings.....	14
Expanding AHC Missions	14
Social Justice.....	16
Readmission and Health Care Utilization.....	17
Impacts on the Future of AHC Education.....	20

Models and Case Studies for Community Engagement	21
Financial Investments of AHCs Towards Community Benefits.....	26
Discussion	27
Gaps in Current Literature	28
CHAPTER 3: METHODOLOGY	30
Study Design and Methods	30
Aim 1	30
Data Sources	33
Data Collection	33
Data Analysis	34
Aims 2 and 3	34
Data Sources	35
Measures	36
Data Analysis Plan.....	37
Delimitations.....	38
Study Limitations.....	39
IRB Considerations/Confidentiality Issues.....	40
CHAPTER 4: FINDINGS	41
Aim 1 Findings	41
Community Benefit Reports	41
Form 990.....	45
Aims 2 and 3 Findings	48
Key Findings in Response to the Research Questions and Aims	49

Summary of Topics and Sub-Topics.....	50
Challenges.....	53
Enablers.....	54
Thematic Findings on SDOH and AHCs in NC	55
Summary of Findings.....	58
CHAPTER 5: DISCUSSION OF STUDY FINDINGS	60
Funding Sources.....	60
Community Partnerships.....	61
Regulatory Agencies.....	62
Health Care Paradigm Shift	64
SODH Key Areas.....	65
Recommendation for NC AHCs	67
Limitations	69
Recommendations.....	70
CHAPTER 6: PLAN FOR CHANGE	71
Create a Sense of Urgency	73
Build a Guiding Coalition.....	73
Form Strategic Vision and Initiatives	74
Enlist a Volunteer Army	75
Enable Action by Removing Barriers	75
Generate Short-Term Wins.....	76
Sustain Acceleration	76
Institute Change	77

Personal Actions to Drive Change.....	79
Conclusion	80
APPENDIX A: TABLE 1.....	82
APPENDIX B: HOSPITAL FORMS	89
APPENDIX C: KEY INFORMANT INTERVIEW GUIDE	99
APPENDIX D: INFORMANT INVITATION EMAIL FOR INTERVIEW	104
APPENDIX E: CODEBOOK	105
REFERENCES	111

LIST OF FIGURES

Figure 1 - Academic Health Centers and the Social Determinants of Health	11
Figure 2 - The 8-Step Process to leading Change – Kotter.....	72

LIST OF TABLES

Table 2 - Summary of Community Benefit Reports for AHCs in North Carolina 2019	42
Table 3 - Itemized benefits as % of Total Community Benefit for AHCs in North Carolina 2019	42
Table 4 - Summary of Community Benefit reports for AHCs in North Carolina 2019 compared to previous year in \$ and % change	44
Table 5 - Summary of Key Informants	49
Table 6 - Summary of a priori topics based on Healthy People 2020	50
Table 7 - Summary of emerging topics based on Key Informant Interviews	51
Table 8 - Summary of SDOH Initiatives	65
Table 9 - Summary of Plan for Change	78
Table 1 - Academic Health Centers and Social Determinants of Health: Community Health Improvement	82

LIST OF ABBREVIATIONS

AAHC	Association of Academic Health Centers
AAMC	Association of American Medical Colleges
ACA	Affordable Care Act
ACO	Accountable Care Organization
ADI	Area Deprivation Index
AHCs	Academic Health Centers
APP	Advanced Practice Provider
BEC	Benefits Enrollment Counseling
BPCI	Bundled Payments for Care Improvement
CBOs	Community Based Organizations
CCP	Complex Child Program
CE	Community Engagement
CEnR	Community-Engaged Research
CE-KFC	Community Engagement Key Function Committee
CHA	Catholic Health Association of the United States
CHCs	Community Health Centers
CHNA	Community Health Needs Assessment
CHOIF	Charlotte Housing Opportunity Investment Fund
CMS	Centers for Medicare & Medicaid Services
CONNECT	Community-Based Organizations Neighborhood Network: Enhancing Capacity Together
CTSA	Clinical and Translational Science Awards

DUH	Duke University Hospital
ED	Emergency Department
EDAC	Excess Days in Acute Care
EHR	Electronic Health Record
EP	Elements of performance
ESRD	End-Stage Renal Disease
FFS	Fee for Service
FQHC	Federally Qualified Health Center
IHI	Institute for Healthcare Improvement
JFU	Just for Us
J-CHIP	The Johns Hopkins Community Partnership
JHHS	Johns Hopkins Health System
KI	Key Informants
KFC	Key Function Committee
LATCH	Local Access to Coordinated Healthcare
LISC	Local Initiatives Support Corporation
MSSP	Medicare Shared Savings Program
NCDHHS	North Carolina Department of Health and Human Services
NC FIT	Formerly Incarcerated Transition Program
NCHA	North Carolina Healthcare Association
NHSC	National Health Service Corps
PADC	Project Access of Durham County
PPE	Personal Protective Equipment

RCT	Randomized Control Trial
SDOH	Social Determinants of Health
SOAR	SSI/SSDI Outreach, Access, and Recovery
SSDI	Social Security Disability Insurance
SSI	Supplemental Security Income
UNC	University of North Carolina

CHAPTER 1: INTRODUCTION AND BACKGROUND

Academic Health Centers (AHCs) have traditionally focused on their three core missions of patient care, research, and education.^{1,2} Patient care at AHCs has generally focused on tertiary care, innovation and specialization.¹ The current changing health care environment requires AHCs to continue to offer specialized tertiary care services to individual patients while also broadening their focus to improve the health of populations.³ The Affordable Care Act (ACA)⁴ in 2010 brought new interest to the community benefits that tax-exempt hospitals provide to the communities they serve. Policymakers predicted that the ACA would result in a decrease in charity care and a shift towards activities improving community health, although this did not necessarily materialize.⁵ There is also an increased interest among AHCs to directly address the social determinants of health (SDOH) as they are viewed as an important factor to the overall health of a population.⁶ For AHCs to adapt to this shifting paradigm, they will need to identify innovative ways to improve the health of their communities and make population health part of their new core mission³ particularly by addressing SDOH.^{2,7}

Background

By some measures, clinical care accounts for only 20% of overall health; the majority of overall health is accounted for by health behaviors such as tobacco use and diet and exercise

(30%), social and economic factors, such as education and employment (40%) and the physical environment such as air quality and housing (10%).^{8,9}

SDOH are the conditions in the environments where people live, learn, work, worship, and play, and they impact a wide range of health risks and quality of life outcomes.^{10,11} Policies and interventions focused on SDOH can positively influence changes in individual behavior that can potentially improve population health.¹⁰ The SDOH definition using the framework developed by Healthy People 2020 is: a “place-based organizing framework, reflecting five key areas of social determinants of health (SDOH)...these five key areas (determinants) include: Economic Stability, Education, Social and Community Context, Health and Health Care and Neighborhood and Built Environment.”¹²

SDOH are related to, but not necessarily the direct cause of, health disparities and inequalities. Braveman et al., defined health equity as “the absence of systematic disparities in health (or in the major social determinants of health) between groups with different levels of underlying social advantage/disadvantage.”¹³ Evidence suggests that policies addressing SDOH have a good likelihood of decreasing health inequity and improve the health of individuals.¹⁴ Disparities in health access, and quality of care are related to individual, social, and environmental factors that impact individuals’ ability to engage in behaviors that promote population health.¹⁵ Reducing health disparities by better understanding and researching SDOH is supported by the National Institute on Minority Health and Health Disparities in their Research Framework published in 2017.¹⁶ Directly engaging communities and addressing SDOH, may reduce health disparities and improve the health of the population.^{2,17} Gottlieb et al., caution, however, that there are also some inherent risks with integrating medical care and social health with possible

unintended consequences related to benefits eligibility, risk prediction modeling, and advances in precision medicine that may worsen inequity.¹⁸

Many not-for-profit health care organizations have community benefit programs to improve the communities they serve, that also serve as the basis for their tax-exempt status.¹⁹ Community benefits as defined by the Catholic Health Association of the United States (CHA) and its partners are “programs or activities that provide treatment or promote health and healing in response to identified community health needs and meet at least one of these community benefit objectives: improve access to health services, enhance public health, advance increased general knowledge or relieve the burden of government to improve health.”²⁰ In 2008, the Internal Revenue service redesigned the Form 990 (Appendix A) and added a section H that includes various categories that may qualify as community benefits. Community building activities (e.g., physical improvements and housing, economic development, and environmental improvements) are important contributors to community health improvement.²¹ The ACA added three types of community benefit requirements in Section 9007. Community Health Needs Assessments (CHNA) that identify key health needs and issues, are required at least every three years and hospitals are required to report on community benefits. Written financial policies for financial assistance and emergency care policies are also required. Lastly the ACA limits charges, billing, and collection activities for uninsured individuals.⁴ These provisions in the ACA encouraged tax-exempt hospitals to invest in community health benefits. However, average spending for all community benefits only increased from 7.6% in 2010 to 8.1% in 2014. Most of the spending went to unreimbursed patient care such as charity care rather than towards community health improvement.²²

An AHC as defined by the Association of Academic Health Centers (AAHC) “encompasses all the health-related components of universities, including their health professions

schools, patient care operations, and research enterprise. Thus, an academic health center consists of an allopathic or osteopathic medical school; one or more other health profession schools or programs such as Allied Health, Dentistry, Graduate Studies, Nursing, Pharmacy, Public Health, Veterinary Medicine and one or more owned or affiliated teaching hospitals or health systems.”²³ There is increasing interest among AHCs to broaden their traditional approach to health care by more directly addressing SDOH that improve individual and population health. Notably, the AAHC has held annual conferences as part of a SDOH Initiative and plans to create an ongoing collaborative focusing exclusively on the SDOH.⁶

The expected ACA expansion in health insurance coverage initiated a renewed interest in expanding primary care. The ACA included provisions that would strengthen and expand Community Health Centers (CHCs) supported by providers through the National Health Service Corps (NHSC). The AAHC conducted a survey of its member AHCs in 2009 to learn more about the evolving partnerships between AHCs and CHCs. Enhancing patient care, education opportunities and population based and translational research opportunities, were identified as benefits of this partnership. Contending with federal requirements and lack of staffing and financial support proved to be impediments.²⁴ As early as 2012 Oliver et al. proposed a new partnership between CHCs and AHCs called the Community Health Center and Academic Medical Partnership (CHAMP). This proposed strategy would expand primary care training while improving the communities served by AHCs.²⁵ The debate about why and how AHCs should contribute to improving the health of their communities has continued during the last decade.

Healthcare reform is shifting the health care landscape, from volume and quantity to a focus on value and quality for the patient, but AHCs are not yet aligned with this shift. There are

both challenges and opportunities associated with the 1) underpinning missions of AHCs, 2) reimbursement methods and models, and 3) community characteristics affecting community health.

1) AHC Missions

The traditional AHC missions of clinical care, research, and education present a challenge to addressing SDOH. These long-established missions need to be reexamined within this new healthcare context that prioritizes value, if AHCs are going to be optimally organized to address SDOH. Smitherman et al. propose a new conceptual framework for AHCs that adds a fourth mission of “social accountability,”⁷ a concept introduced in medical schools over twenty years ago. Social accountability has been defined as “the obligation of medical schools to direct their education, research, and service activities towards addressing the priority health concerns of the community, region, and/or nation they have a mandate to serve. The priority health concerns are to be identified jointly by governments, healthcare organizations, health professionals and the public.”²⁶ Smitherman et al. claim that social accountability for AHCs requires engagement at three levels: identifying the primary health needs and challenges, adapting the institution’s programs in education, research, and clinical care to address these health needs and challenges, and assessing the impact of the institution’s efforts. In addition to an ethical and moral obligation, the authors advocate for this mission because of the current breadth and influence of AHCs, the publicly funded benefits that AHCs enjoy, and community and public sector contributions to AHCs. The authors conclude that addressing SDOH prioritizes the primary driver of poor health outcomes in the United States, necessitating this proposed new model and mission.⁷

2) Value-Based Models

The second major challenge that AHCs are currently facing is a shift in reimbursement models spurred by healthcare reform, specifically the passage of the ACA in 2010.⁴ The Centers for Medicare & Medicaid Services (CMS), is committed to promoting accountability and quality of care at a reduced cost through initiatives such as the Medicare Shared Savings Program (MSSP). CMS is also evaluating other models such as the Accountable Health Communities Model, which seeks to reduce health care utilization and costs by addressing social needs such as unstable housing for Medicare and Medicaid beneficiaries through screening, referrals and community navigation services.²⁷ These programs shift reimbursement from a volume-based approach to value-based care that focuses on quality, the patient experience, and population health.²⁸

AHCs face challenges as they adapt to these shifts. The changing health care environment requires that they put more emphasis on population health while still offering specialized services. Traditionally, clinical care at AHCs is episode-based with an acute care emphasis and is not focused on the long-term view of improving the health of their communities or a population. AHCs are now facing a transition where they need to evolve and broaden their focus to population health improvement while preserving their ability to perform tertiary and quaternary care. Therefore, two models of care need to be currently maintained: (1) the prevailing Fee for Service model (FFS) emphasizes volume and increasing the number of patients and procedures to increase reimbursement and (2) the value-based care model reimburses based on quality and outcomes. Maintaining these two competing models requires a delicate balance that is difficult, if not impossible, to sustain. The economic realities of an increased focus on quality and outcome measures, such as 30-day readmission complications and excess days in acute care (EDAC)²⁹ has

necessitated AHCs to reexamine their way of doing business and broaden their focus towards primary care and population health and makes the prevailing FFS model unsustainable.³⁰ As the transformation towards quality and value-based care intensifies, the incentive to increase the number of patients and procedures performed continues to decline. In a predominantly value-based system, avoiding hospitalization and unnecessary procedures may result in health care savings and potentially improve outcomes and patient experience. AHCs may well prove to be a catalyst for promoting value-based care because they have resources in care management, health information technologies, and community partnerships that can be leveraged to meet social needs. AHCs also have expertise in focusing on the patient populations with the most complex needs, and building interdisciplinary teams to engage patients, families and caregivers.³¹

Value-based models are emerging as the predominant model of care, at a faster pace than anticipated. According to a survey by ORC International in 2018, pure FFS reimbursement, which was at over 50% in 2016, is fading faster than predicted in past studies. It now accounts for 37.2% of reimbursement and is projected to dip below 26% by 2021.³²

The models of value-based care can be categorized as bundled payments, Accountable Care Organizations (ACOs), and other types of risk-based contracts. In bundled payments models, the total cost of an episode of care is predetermined. Participants in such models share in any losses or savings that result from the difference between the actual and targeted costs. This model was implemented by CMS as part of the Bundled Payments for Care Improvement (BPCI) Initiative and involves a redesign of care delivery focused on interprofessional collaboration.^{33,34} ACOs are organizations comprised of multiple health providers within health systems that coordinate care to populations of patients. The goal of ACOs is to improve the quality of care and lower costs. ACOs now serve many beneficiaries through Medicare, Medicaid and various types

of commercial contracts involving insurers, hospitals, and alliances of physicians and other providers. CMS led this charge through the Pioneer ACO model and the Medicare Shared Savings Program (MSSP) in 2012 and more recently the Next Generation ACO model and the Comprehensive End-Stage Renal Disease (ESRD) Care Model in 2016.³⁵ Other risk-based payer contract models with variable levels of risk exist, including patient-centered medical homes (PCMH) and risk-based contracting between insurers and large health systems or integrated delivery networks. Employers are also entering into value-based contracts with insurers to improve the care to their employees and lower costs. Full capitation, in which providers receive a capped monthly payment per person regardless of whether a person seeks care or not, is at the extreme end of risk compared to other risk-based contracts.

As financial risk continues to be shared with health delivery systems under the various types of value-based care models described above, AHCs will need to increase the resources it commits to the longitudinal care of patients as part of “lifetime health management” from prevention to end of life services.³⁶ This creates a new emphasis on population health improvement in which AHCs become more accountable to improving the health of their community or population. AHCs will need to create new partnerships, including primary care networks, if they are to adapt to the changing environment. Conflicting incentives amid the transformation pose a significant challenge. AHCs will need to continue to invest in high-end technology to provide tertiary and quaternary care, while simultaneously developing primary care networks and incentivizing providers to provide value-based care.³⁷ As AHCs continue to evolve toward value-based care, the challenges of integrating population health improvement into their traditional roles of clinical care, education and research will need to be resolved.

3) Community Characteristics

Community characteristics such as affordable housing, food deserts, unpolluted environment and public transportation affect the health of individuals, the community, and the population. Community health is defined as “a multi-sector and multi-disciplinary collaborative enterprise that uses public health science, evidence-based strategies, and other approaches to engage and work with communities, in a culturally appropriate manner, to optimize the health and quality of life of all persons who live, work, or are otherwise active in a defined community or communities.”³⁸ Population health for this study will be defined as “the health outcomes of a group of individuals, including the distribution of such outcomes within the group.”³⁹ “Population health” will be viewed as a concept of health and “population health improvement” as focused on community health as used by the National Academies of Sciences in their recent Roundtable on Population Health Improvement: “product of multiple determinants of health, including medical care, public health, genetics, behaviors, social factors, and environmental factors.”⁴⁰

If AHCs hope to be competitive in a value-based world, they may need to move towards creating “academic-community health systems” where communities and AHCs become interdependent. To have a stronger impact on improving long-term SDOH, community involvement and integration at all levels and curricula may be necessary, including teaching future health care professionals about SDOH. Furthermore, collaboration with communities and AHCs is critical to develop and implement sustainable, community-specific interventions that address SDOH. Finally, AHCs must refocus some of their research towards evaluating community-based interventions that specifically address SDOH and reduce health disparities.^{2,17} To achieve increasing social accountability, AHCs will need to develop partnerships with key community stakeholders

and measure the impact of their interventions on meeting the needs of the community. AHCs receive significant public funding, which presents an obligation and opportunity to move towards increasing social accountability. They have specialized services capabilities and often serve as a safety-net for their community when providing charity care.⁷ Consuelo et al. argue for AHCs to embrace community engagement and integration, to move towards a cultural shift of improving health. They argue for commitment from AHC leadership and a centralized infrastructure to truly engage the community engagement in their core missions of research, medical education and clinical care.⁴¹

This study will extend the literature by exploring opportunities and challenges faced by AHCs when addressing SDOH to improve the health of patients and their communities. Rather than focusing only on defined patient populations within their delivery systems (such as ACOs), this study will focus on the opportunities and challenges AHCs face when contributing to broad initiatives and partnerships particularly to improve the health of their communities.

Conceptual Framework

The conceptual model (Figure 1) for this study is adapted from the conceptual model of SDOH from Healthy People 2020. SDOH are divided into five primary areas of economic stability, education, health and health care, social and community context and neighborhood and built environment. The traditional core missions of clinical care, research and education insufficiently address SDOH. By adding a fourth core mission of social responsibility, and taking an active role in addressing SDOH, AHCs can improve the health of patients and their communities.



Figure 1 Adapted from <https://www.healthypeople.gov/2020/topics-objectives/topic/social-determinants-of-health> and Smitherman, H. C., Baker, R. S., & Wilson, M. R. (2019). Socially accountable academic health centers: pursuing a quadripartite mission. *Academic Medicine*, 94(2), 176–181.

Aims and Research Question

The research question for this study is “What are the opportunities and challenges for AHCs addressing SDOH to improve the health of patients and their communities?”

The aims of this study are.

- Aim 1: To evaluate the contributions of AHCs in North Carolina to activities benefiting their community.
- *Method – Analysis of community benefits reports and Form 990 Schedule H when available.*
- Aim 2: To explore how AHCs in North Carolina are partnering with local and national organizations to improve the health of patients and their communities.

Method – Key informant interviews with AHC leaders.

- Aim 3: To identify challenges that AHCs in North Carolina are trying to overcome and enablers to improve the health of patients and their communities.

Method – Key informant interviews with AHCs leaders

CHAPTER 2: LITERATURE REVIEW

Information Sources

Studies reviewed were identified through searching three primary databases: PubMed, SCOPUS, and Web of Science. The specific search terms used were:

"Academic medical center" OR "Academic health system" OR "Academic health centers" OR "Academic Medical System"

AND

"Social determinants of health" OR "Social determinant of health" OR "determinants of health" OR "SDOH" OR "SDoH" OR "social needs" OR "social risk"

AND

"Community health" OR "community health building" OR "community health improvement" OR "community"

The search included articles published between January 1, 2010 (the year the ACA was passed into law) and September 12 of 2020. Only included articles related to the US health care system and written in English were included. References that were descriptive, causal, quantitative, qualitative, case study, or case report and gray literature were included. Opinion pieces, viewpoints and editorials were also reviewed. Several other references were identified using the reference lists from articles that were included in the review.

Reference Selection

All references were compiled into Covidence for screening of titles and abstracts by the reviewer. Duplicates were removed using Covidence, and the reviewer also confirmed deduplication manually. The eligibility of references was performed by one reviewer due to time and logistical constraints. The initial screening was based on the title and abstract; the full text of potentially eligible were reviewed.

Summary of Findings

The review identified opinion pieces and editorials arguing for expanding AHC missions and social justice, studies exploring readmission and ED utilization, impacts on the future of AHC education, models, and case studies for addressing SDOH and community engagement, and lastly a look at the financial investments of tax-exempt hospitals including AHCs towards community benefits. Table 1 in Appendix A includes more detail on each article referenced in this Chapter.

Expanding AHC Missions

Park et al. argue that the missions of education, research and clinical care should be focused on social determinants to create partnerships with their communities. These “academic-community health systems” would advance health equity through four primary strategies: co-creation with their communities for more inclusive workforce recruitment, implementing curricula

that emphasize how SDOH disproportionately affect health outcomes, offer opportunities for health care professionals to partner with communities to address health inequities and finally develop and evaluate community based research agendas that attempt to address health inequities.² Karpf agrees that AHCs need to play a significant role in addressing SDOH but views their role as more limited. The author argues that providing affordable and comprehensive health coverage for all is a necessary first step and federal, state, and local governments have the necessary resources to address poverty and the roots of the SDOH. Wartman et al., argue for a nation-wide movement of AHCs to address SDOH collectively. Several barriers to accomplishing this are discussed. The “guild mentality” of health professions promotes competition instead of collaboration and existing AHC structures including promotion, tenure and fragmentation currently impede the ability of AHCs to collectively address SDOH. Regulations and accreditation and misaligned financial incentives of the US health care system can also serve as barriers. The authors propose a new model where multi-disciplinary teams in the areas of education, research and clinical care across departments have a specific area of focus. The model also suggests the creation of a compact between AHCs and community leaders, that establishes rules and guidelines for all interactions between an AHC and their community. The compact would include community-based participatory research, involve the community in developing a new curriculum, expand training sites and improve care to better address the SDOH.⁴² Perman et al. similarly argue for “infusing” SDOH in all aspects of AHC clinical care, research and education through community partnerships and a shift to a disease-prevention approach.⁴³ Community engagement and integration is also viewed as a vital component to help AHCs better understand and address SDOH by Wilkins et al. The authors argue that communities have been engaged in limited ways in the past and the necessary infrastructure to support community partnerships across their research, clinical

and education missions must be created for long term success.⁴¹ In an issue brief on behalf of the AAHC, Knettel makes a business case for AHCs to address environmental, social, and behavioral determinants of health. AHCs have often served as safety-net providers and offer care to low-income patients and those with limited access to care. The fear that addressing SDOH will result in declining revenues is overstated and it is unlikely that there will be a decrease in patients for AHCs, especially in the short term. Demographic shifts and new technologies may well offset any loss of revenue. There are also good arguments to addressing SDOH more broadly, including community recognition and reputation that may well attract more patients. AHCs may also find they would experience a reduction in uncompensated care if SDOH were addressed. Reimbursement policy is also moving towards population and outcome-based care which may also make an important business case for AHCs to address SDOH now to be better prepared for the anticipated post-FFS marketplace.⁴⁴

Social Justice

Social justice and social accountability are themes addressed in some commentaries.^{7,45,46} In a commentary by Schroeder reacting to the Keystone IV conference for Family Medicine, the author comments on the recurring theme for the essential role of social justice as the “moral core” and expressing solidarity with the community. The author cautions that adding this value into practice remains challenging.⁴⁵ Alberti et al. argue that AHCs have an ethical obligation to address issue of social and economic injustice because they threaten the health of the communities that AHCs serve. AHCs have many resources and are often located in underserved communities. AHCs partnering and engaging with their communities can be a crucial way to promote

justice and equity.⁴⁶ Smitherman et al. make the strongest appeal to make social accountability a fourth mission for AHCs, in addition to the traditional missions of education, research and clinical care. The authors propose a conceptual framework with this new quadripartite mission in exchange for the traditional tripartite mission. They believe AHCs have both a mandate and an obligation to add social accountability to their mission. The authors justify this proposed new framework on moral and ethical grounds, the influence and impact of AHCs, resources that AHCs currently have (including infrastructure and expertise), and the fact they are already recipients of public sector investment. Comprehensive and deliberate community engagement that includes metrics to measure the impact of any interventions are an integral part of this proposed perspective.⁷

Readmission and Health Care Utilization

Several studies examined how addressing SDOH affected health care utilization particularly Emergency Department (ED) utilization and readmission within 30 days.^{47,48,49,50}

Wu et al. conducted a cluster randomized control trial (RCT) between 2014 and 2016 aimed at reducing healthcare utilization and increasing referrals between an AHC and local Community Based Organizations (CBOs). The primary aim of the Baltimore Community-Based Organizations Neighborhood Network: Enhancing Capacity Together (CONNECT) study was to strengthen the partnership between the AHC and CBOs to better address patient needs. The primary outcomes measured were emergency department visits and days in the hospital at the Johns Hopkins Health System (JHHS). JHHS partnered with 22 CBOs randomly assigned to the intervention or control group (N=5,255 patients). The intervention consisted of an online tool to refer

clients to community resources and meet and greet sessions for CBO staff with healthcare staff. The intervention did not improve healthcare utilization outcomes but did increase staff knowledge and confidence in local CBOs. Baltimore CONNECT however, became a 501c3 non-profit organization with the mission of addressing SDOH to improve the health of Baltimore City residents.⁴⁷

A pilot quality improvement intervention for 72 patients in an urban AHC (Brigham and Women's Hospital) also used an RCT to study whether an intervention improved care coordination and reduced ED visits and hospitalizations among frequent ED users. The intervention consisted of a community health worker (CHW) who assisted patients with unmet social needs (e.g., coordinating transport and providing food bank information), and a team that developed an interdisciplinary care plan for eligible patients. The intervention group had 35% lower ED visits, 31% fewer admissions, and 15% lower costs than the control group. The results are not statistically significant due to the small sample size; however, this may be a promising approach.⁵¹

Hospital readmission and social risk factors were studied in an observational study by Navathe et al. using physician notes, payer claims and Electronic Health Record (EHR) data. They studied seven social risk factors: tobacco use, alcohol abuse, drug abuse, depression, housing instability, fall risk, and social support in patients with cardiovascular disease. The authors concluded that adding natural language processing to analyze physician notes revealed the seven social risk factors were more prevalent than using administrative data alone. The data from physician notes enable better detection of readmitted patients and may lead to better predictive models in the future.⁴⁸

Zhang et al. conducted a retrospective cohort study of over 12,000 patients on the impact of individual (e.g., sex, race) and community (e.g., unemployment rate, median income) SDOH

on predictive models for 30-day readmission for Medicaid patients, patients who are 65 or older, and/or obese patients. SDOH improved the readmission risk prediction for all three vulnerable subgroups studied but did not meaningfully improve readmission prediction for the general patient population. This may indicate prediction models may be more effective when targeted to more vulnerable patients.⁵²

Graham et al. attempted to differentiate between causal factors for early (0 to 7 days) and late (8 to 30 days) readmissions using a retrospective cohort study design. SDOH that included education level, barriers to learning, and type of insurance were significant predictors of readmissions in the later period.⁴⁹ In a qualitative study using semi-structured interviews to explore causes of early readmission among patients with diabetes, SDOH impeding care (e.g. limited transportation, help with preparing food) were identified.⁵³ Georgiev et al. describe how nurse practitioners in an AHC in Maryland participated in a community paramedicine transitional health support program to work with complex patients after discharge. Community health workers identify eligible patients by identifying SDOH for a vulnerable population and the multidisciplinary team aims to reduce 30-day hospital readmissions, ED visits and overall costs. Preliminary data have shown the potential to reduce costs and improve the quality of care.⁵⁴

Hatef et al. evaluated whether housing at both an individual and community level impacted health care utilization at JHHS. This descriptive analysis used both data extracted from the EHR, and the patient's neighborhood as measured by the Area Deprivation Index (ADI), a composite measure that ranks neighborhoods by their socioeconomic disadvantage. Of more than a million participants, they identified over 50,000 (5.8%) with at least one housing issue. After adjusting for other factors, patients with housing instability had 49% more health care encounters, and patients experiencing homelessness had 34% more encounters. Persons living in the

most disadvantaged neighborhoods had 55% more encounters than those living in the most advantaged neighborhoods. This findings can be used for risk prediction, care coordination, population health management and more focused community-based interventions.⁵⁰

Impacts on the Future of AHC Education.

Gonzalo et al. examine new competencies that medical education must incorporate in both the clinical arena and classrooms to prepare future providers. New health systems science competencies include value-based care, quality improvement, SDOH, population health, informatics and systems thinking. Both current and future faculty may need to develop their teaching skills to incorporate the new competencies. Because many physicians lack these skills and formal training, the authors argue that these new competencies require interdisciplinary collaboration, including non-physician clinicians (e.g., social workers, nurses, nurse practitioners, physician assistants) as well as experts in quality improvement and system leadership. AHCs will also be challenged to find innovative ways to support and compensate all contributing faculty through protected time, compensation, and potentially formal faculty status. These changes may better align the education and clinical mission of AHCs, and ensure that the next generation of physicians can lead system transformation.⁵⁵

Students will also need to learn to work on interprofessional teams that respond to community needs. The Camden Coalition of Healthcare Providers (the Coalition) partnered with the Association of American Medical Colleges (AAMC) in 2014 and launched the Interprofessional Student Hotspotting Learning Collaborative to train students in working as an interprofessional

team. “Hotspotters” are interprofessional teams that engage with communities to care for individuals with complex health and social needs. One of the goals of this collaborative is to increase student awareness of the role of SDOH and their ability to impact those determinants in the future. Ten student teams participated including students from The University of North Carolina (UNC) School of Medicine and Duke University School of Medicine. Students from Virginia Commonwealth University indicated that this program was successful in teaching them how SDOH affect health; the authors recommend a larger study to evaluate the program’s impact on patients and the community.⁵⁶

Models and Case Studies for Community Engagement

Models for community engagement with AHCs are proposed to improve community-based research, improve population health, and decrease costs.^{57,58,59,60} A taxonomy for organizing and discussing community health indicators is proposed by Aguilar-Gaxiola et al.⁵⁷ as part of the Community Engagement Key Function Committee (CE- KFC) for the Clinical and Translational Science Awards (CTSA) Consortium in which more than 60 research institutions participate.⁶¹ The proposed taxonomy proposed, supports activities and research across CTSA using the CTSA Community Engagement Logic Model. The taxonomy starts at the broadest level with determinants of health (e.g., disparities), then progresses to categories of community health indicators (e.g., gender) and finally arrives at specific quantifiable measures (e.g., US Healthy People 2020). Determinants of health include the social, economic, and physical environment and a person’s individual characteristics and behaviors. Community health indicators are characteris-

tics of an individual, population, or environment. Finally, quantifiable measures are standard reference points that can be evaluated. The authors provided examples of how the taxonomy could be used for health indicator projects. The authors advocate for evaluating both the process and impact of community-academic partnerships that hold great potential to improve population health.

Translational medicine through Community Engagement (CE) and Community-Engaged Research (CEnR) can be an important way for AHCs to improve population health and decrease costs. Michener et al. argue that a key challenge is to align the goals of AHC leadership and community stakeholders. The CTSA consortium advances CEnR through the CE-Key Function Committee (KFC) by effectively engaging communities and clinical practices in bidirectional dialogue. The five steps the authors propose are: defining community and identifying partners, learning the etiquette of CE, building a network of CEnR researchers, developing new methodologies, and improving translation and dissemination plans. AHCs funding and tenure and promotion policies may be barriers to this approach; AHCs may need to change their culture to align their goals with their communities to ultimately improve the health of the nation.⁵⁹

Devoe et al. advocate for AHCs to adapt their infrastructure from combating disease alone to also promoting health. They propose fostering community “laboratories” focused on “the science of health”, cultivating community “classrooms” to create learning opportunities in the community and then building partnerships between AHCs and these laboratories and classrooms. The authors argue that this can be accomplished with six primary strategies: strengthening academic primary care; building primary care extension programs; supporting patient-centered medical villages; supporting communities as learning systems for health; investing in communities so they can be a resource for research; and cultivating and strengthening community

classrooms. AHCs will need to make longer term investments in community-academic partnerships to move toward population health. The authors propose creating an Institute for Healthy People based on the pillars of AHCs, community laboratories and community classrooms.⁵⁸

Greatly expanding the role of CHCs by partnering with AHCs represents a potential strategy to serve more Medicaid patients.⁶⁰ The authors argue that a new paradigm powered by mega “Teaching Health Centers” in partnership with AHCs could control costs of Medicaid expansion and increase the likelihood for bipartisan support for the ACA. The success of this model would depend on the ability to increase the primary care provider pipeline and access to care by adding more CHC sites. Currently, most CHCs employ physicians in just one of the primary care specialties, most often family medicine. CHCs would be staffed by more than the traditional family medicine provider by adding the specialties of internal medicine and pediatrics. New community facilities would include nurse practitioners and physician assistants to increase access to care and decrease costs. Other aspects of this model include interprofessional education for physicians and trainees and increasing community health services research.

The review of the literature revealed that there are some examples of AHCs that integrate SDOH into their clinical practice while partnering with their communities. The primary focus is for patients within their health care delivery systems. The Johns Hopkins Community Partnership (J-CHIP) is an example of an AHC-based comprehensive program that seeks to improve care across different settings by addressing SDOH. The AHC partnered with two CBOs and five skilled nursing facilities to facilitate transitions to post-acute care.⁶² The University of Toledo College of Medicine is partnering with ProMedica to support a large -scale economic development to address the SDOH through a 50-year, \$2.5 billion relationship. They have established a series of short-term goals including neighborhood redevelopment plans, the development of the

ProMedica National Center for Social Determinants of Health Research and the recruitment of new faculty to lead this transformation.⁶³ The University of New Mexico works with the Health Extension Regional Officers to address SDOH and engage in community-based research. The Oregon Rural Practice Research Network uses four pillars to engage the community: community engagement, practice transformation, research, and education and serves as a link between researchers, health policy experts, educators and the health extension workforce.⁶⁴

Several recent studies examined strategies in which AHCs partnered with CBOs to improve the health of their communities. A study of four AHCs in Massachusetts evaluated an innovative community-driven intervention to reduce disparities in breast cancer treatment using three evidence-based strategies: patient navigation services, a shared patient registry across participating AHCs, and a web based SDOH platform to identify at risk patients and direct them to available resources.⁶⁵ New York University Langone Health partnered with local CBOs to oversee their CHNA. Through this partnership, they designed, implemented, and evaluated several community health improvement projects. Two such programs are ParentCorps (supports high-risk parents and children in early childhood education) and Tobacco Free Community (assists low-income adults with cessation therapies). The authors recommend supporting building capacity for CBOs, integrating programs into operational flow, and leveraging resources to increase the impact of community health improvement activities.⁶⁶ Researchers at the University of Chicago evaluated the impact of connecting patients to community-based resources using an automated resource called CommunityRx. Participants gained knowledge about common resources in the community related to smoking cessation and weight loss; higher knowledge was associated with higher resource use which may improve behaviors.⁶⁷ Wynn et al. performed a quality im-

provement initiative where patients identified as food insecure were linked with community resources and provided with a bag of food at discharge. Food insecurity at this urban AHC was identified in 10% of the screened patients. The authors concluded their pilot study was feasible and sustainable and could serve as a template to address other SDOH.⁶⁸

New York Presbyterian Hospital at Columbia University combined universal SDOH and clinical screening in four outpatient primary sites. The AHC screening tool for SDOH was completed on tablets via NowPow, a community referral platform that screens for SDOH to address population health and gaps in community care, assigns risk levels, and refers patients to resources based on their risk.⁶⁹ NowPow screens for food insecurity, housing insecurity and transportation needs.⁷⁰ Fiori et al. implemented a social needs screening program at a pediatric clinic in an urban AHC. The Community Linkage to Care program created a partnership between the AHC and community partners including a Federally Qualified Health Center (FQHC) to create an integrated social needs screening and outreach model. A screening workflow, a referral workflow and systematic patient follow up were established. Provider champions and administrative liaisons were key players in this program while progress was monitored with performance improvement activities. The need for childcare, housing quality and food were the most frequent social needs.⁷¹

Through Carolina Health Net, UNC partnered with Piedmont Health Services, a community health center (CHC) to help connect uninsured patients with primary care with the goal of avoiding costly ED visits. Piedmont is set up in a way to seamlessly transition patients to a Medicaid managed care program called Community Care of North Carolina. Key interventions include providing medical homes, transitioning patients discharged from the ED and inpatient settings, coordinating subspecialty care, and assisting the uninsured with Medicaid applications.⁷²

Financial Investments of AHCs Towards Community Benefits

Literature on investments by AHCs specifically in SDOH is limited. Prior to and in the early years after the ACA became law, hospitals invested little in SDOH. Rosenbaum et al. estimate that tax-exempt hospitals spent \$24.6 billion on community benefit in 2011 (up from 12.6 billion in 2002).⁷³ Overall hospitals spent about 8% of their operating expenses on community benefits with the majority on charity care. Merely five percent of community benefit spending is devoted to community health improvements, only a small portion of which is dedicated to SDOH.^{74,75}

Horwitz et al. searched all public announcements of new programs by health systems involving SDOH investments from January 2017 to November 2019. They adapted the Healthy People 2020 model excluding programs solely in the health domain. They found large investments totaling \$2.5 billion by health systems (none of which were for-profit) towards SDOH: several programs were multifaceted with \$1.6 billion in 52 programs allocated to housing; \$1.1 billion in 28 programs focused on employment; \$476.4 million in 14 programs for education; \$294.2 million in 25 programs for food security; \$253.1 million in 13 programs for social and community context; and \$32 million in six programs for transportation. Programs were taking place in 30 states; only two programs committing \$12 million were attributed to North Carolina.⁷⁶ Fos et al. examined the distribution and variability in community benefit spending among North Carolina's 53 tax-exempt hospitals in 2012. The main outcome measure was dollar expenditures as a percentage of operating expenses for community benefit spending. Using published community benefit reports from the North Carolina Hospital Association, community benefit spending was \$2.6 billion, 85% of which was directed at patient financial assistance. Overall,

only 0.7% was allocated to community building activities that address SDOH. This data is similar to that observed by US tax exempt hospitals overall.⁷⁷

Discussion

There is a growing consensus that AHCs should add SDOH to their traditional missions of patient care, education, and research. Some authors advocate adding SDOH to the three main missions while others argue for a more fundamental change that emphasizes SDOH to improve the population health of their communities. While some authors cite ethical and social obligations, others make a business case in response to the current transformation in our healthcare system. Successfully addressing SDOH in a community may decrease the amount of uncompensated care that many AHCs provide. Although AHCs are increasing their attention towards SDOH in a clinical setting, they remain more focused on managing the patients within their delivery systems through their ACOs and initiatives such as BPCI.

SDOH are also playing a more important role in the education of the next generation of health professionals, but many challenges remain. The structure of AHCs including tenure and promotion, and siloed academic departments are some of the identified barriers that will need to be addressed to promote more collaboration. The limited number of faculty who are trained in important new skills and competencies can also limit the quality of education for learners. Academic-community partnerships will need to grow to offer the right type of experiences for students to appreciate the importance of SDOH and population health.

Although there is a growing body of research studying interventions related to SDOH, there is limited evaluation of these interventions. In a systematic review of interventions on social and economic needs at the patient level from January 2000 to December 2014, Gottlieb et al. found a variety of research methods and designs. Many of the interventions are often evaluated for process such as identifying unmet social needs rather than healthcare utilization outcomes. The authors questioned the overall evaluation of interventions and rated most of the studies as low quality.⁷⁸ Overall AHCs have not yet shifted their attention to significantly invest in improving their communities by addressing SDOH. Most community benefits are going towards patient financial assistance including charity care rather than community building activities.

Gaps in Current Literature

This literature review reveals gaps in current literature as it relates to SDOH and AHCs. As value-based care is better defined, and SDOH are incorporated into new models, a more complete assessment of how AHCs are addressing SDOH is needed. Lessons learned from evaluations and success stories can be shared to create best practices and identify pitfalls to avoid. North Carolina is an excellent place to conduct this research, because it is a state that is greatly impacted with unmet social needs. More than 1.2 million North Carolinians cannot find affordable housing and the state also has the 8th highest rate of food insecurity in the US.⁷⁹ It is also one of only 11 states that has not yet expanded Medicaid and one of its primary challenges is a high percentage of uninsured population.^{80,81} Little information about the AHCs in North Carolina included in this study is currently available in the literature. This study explores how AHCs in

North Carolina are partnering with local and national organizations to improve the health of patients and their communities and identify challenges that AHCs in North Carolina are trying to overcome to improve the health of patients and their communities. It also evaluates the contributions of AHCs in North Carolina to activities benefiting their community.

CHAPTER 3: METHODOLOGY

Study Design and Methods

A sequential mixed-methods methods design was conducted in the five AHCs in North Carolina: UNC Health, Duke University Health System, Vidant Health (now ECU Health), Wake Forest Baptist Health (now Atrium Health Wake Forest Baptist), and Atrium Health (now Advocate Health). Within these health systems, UNC Medical Center, Duke University Hospital, Vidant Medical Center (now ECU Health Medical Center), Wake Forest Baptist Medical Center (now Atrium Health Wake Forest Baptist) and Carolina's Medical Center are the five academic medical center teaching hospitals in the state of North Carolina. Each of the five AHCs has several other smaller hospitals within the larger systems. Aim 1 used data from Community benefit reports and Form 990 Part H for the flagship medical centers when available. Aims 2 and 3 used Key Informant (KI) interviews. Aim 1 was completed before Aims 2 and 3.

Aim 1

The quantitative part of the study addressed Aim 1 of the study using descriptive statistics to evaluate how the selected AHCs are benefiting their communities.

- Aim 1: To evaluate the contributions of AHCs in North Carolina to activities benefiting their community.

Financial data from community benefits reports and the Form 990 Schedule H (when available) were analyzed for the five AHCs in North Carolina. Hospitals in North Carolina voluntarily submit their community benefit reports to the North Carolina Healthcare Association (NCHA). Community benefits reports include a list of Community benefits (from A through R) and Bad debt Costs (bullet S)⁸²:

- A. Estimated Costs of Treating Charity Care patients.
- B. Estimated unreimbursed costs of treating Medicare patients.
- C. Includes an adjustment in this period's Medicare revenues for extraordinary adjustments¹ of.
- D. Without this Medicare adjustment, Medicare losses would have been $(B + C)$.
- E. Estimated unreimbursed costs of treating Medicaid patients.
- F. Includes an adjustment in this period's Medicaid revenues for extraordinary adjustments of.
- G. Without this Medicaid adjustment, Medicaid losses would have been $(E + F)$.
- H. Estimated unreimbursed costs of treating patients from other means-tested government programs.
- I. Includes an adjustment in this period's other means-tested government program revenues for extraordinary adjustments of.
- J. Without this adjustment, other means-tested gov. program losses would have been $(H + I)$.
- K. Community health improvement services & community benefit operations.

L. Health professions education.

M. Subsidized health services.

N. Research costs.

O. Cash and in-kind contributions to community groups.

P. Community Building Activities.

Q. Total Community Benefits¹ with Settlements and Extraordinary Adjustments (A + B + E + H + K + L + M + N + O + P).

R. Total Community Benefits¹ without Settlements and Extraordinary Adjustments (A + D + G + J + K + L + M + N + O + P).

Bad Debt Costs

S. Estimated costs of treating bad debt patients.

Form 990 Schedule H has six parts.

- 1) Charity Care and Certain Other Community Benefits at Cost.
- 2) Community Building Activities.
- 3) Bad Debt, Medicare, and Collection Practices.
- 4) Management Companies and Joint Ventures.
- 5) Facility Information.
- 6) Supplemental Information including promotion of community health.

Community-building activities (Appendix B, Part II) directly addresses all SDOH as described in Healthy People 2020 including economic stability, education, health and health care, social and community context and neighborhood and built environment.

- 1) Physical improvements and housing (neighborhood and built environment).

- 2) Economic development (economic stability).
- 3) Community support (social and community context).
- 4) Environmental improvements (neighborhood and built environment).
- 5) Leadership development and training for community members (education).
- 6) Coalition building (social and community context).
- 7) Community health improvement advocacy (health and health care).
- 8) Workforce development (education).
- 9) Other (Variable).

Data Sources

This study focused on the five North Carolina AHCs. The Community Benefit Reports for fiscal year 2018 were accessed from the NCHA website.⁸³ Form 990 Schedule H information was accessed via the GuideStar website where available.⁸⁴ The UNC Medical Center and Carolina's Medical Center are not required to file a Form 990 because in addition to a not-for profit hospitals they are also instrumentalities of states or political subdivisions. The Form 990 was available for the other three AHCs in the study.

Data Collection

Financial data for the various expenses reported in the community benefits for 2018 were collected from the NCHA website and tabulated in Excel. Financial data were also collected from the Form 990 Schedule H for 2018 from the GuideStar website where available.

Data Analysis Plan

Expenses were analyzed and compared within and across the AHCs using data from the community benefit reports. The percentage of the different types of benefits provided were calculated. The percentage of community benefit expenses as part of the total operating expenses were calculated for each hospital. The dollar amount and percentage changes in community benefits from the previous year of available data for each AHC were calculated.

Aims 2 and 3

These aims explored opportunities and challenges associated with efforts to improve the health of patients and their communities by improving SDOH. Opportunities and challenges that NC AHCs are facing when attempting to meet SDOH to improve the health of their communities were identified.

- Aim 2: To explore how AHCs are partnering with local and national organizations and finding innovative ways to improve the health of patients and their communities.
- Aim 3: To identify challenges that AHCs are trying to overcome and enablers to improve the health of patients and their communities.

Four KI interviews were completed at each of the five AHCs using the Key Informant Guide (Appendix C). The Key Informant Guide was first pilot tested with an administrator outside of the five AHCs included in this study. The administrator was identified through the author's personal connections. The pilot interview findings were not included in the results. The Key Informant Guide was slightly adjusted to improve content and flow, based on the pilot prior

to the actual KI interviews included in the study. Probes were added as needed for each question to encourage deeper thinking. The KI interviews at the selected AHCs provided qualitative data to learn more about the current efforts each organization is undertaking, regarding addressing the SDOH to improve the health of patients and their communities.

Data Sources

Administrators at the manager level or above working in population health (e.g., director of population health) were contacted by email (Appendix D). Administrators were identified by their described role from the AHC websites or personal connections. Other connections through the dissertation committee members were also used to determine initial contacts for the study as appropriate KIs. The email included a standardized introduction to the purpose of the research, expectations for the interview, and strategies for ensuring confidentiality of responses. The email explained that they are not obligated to participate. A second email was followed one week after the first if no response was received and the third and final attempt was made to the potential interviewee one week after that. The email requesting participation included a note that if the recipient does not feel they can provide adequate answers to the questions, they can provide alternative contacts that they know will be able to do so.

When participants agreed to be interviewed, an appointment was scheduled at a time convenient to them. All the interviews were conducted through video conferencing via WebEx or Zoom. Consent was reviewed orally by the principal investigator (including obtaining permission to record the session), and the participant was invited to ask any questions. All the potential interviewees who accepted the invitation to participate consented to participate in the study and

were interviewed. The sessions were recorded using a digital audio recorder and the recordings were transcribed for analysis. No monetary or non-monetary incentives were offered to participate in the study. To maintain confidentiality, each subject was given a numeric identifier so their specific comments could not be linked to the data. Given the small sample size, there may be instances where the respondents' identities are clear based on their examples or perspectives. In these situations, every effort was made to convey the information without revealing identifying details.

Snowball sampling was used to identify KIs. After the initial session, the first and subsequent interviewees at each site were asked to recommend other people who may be interested in participating based on their perspectives and/or experience addressing SDOH to improve the health of patients and their community until four KIs were interviewed for each site.

Measures

The KI Interview questions were derived from the SDOH framework developed by the Healthy People 2020.¹⁰ The guide reflected five key areas of SDOH: Economic Stability, Education, Social and Community Context, Health and Health Care and Neighborhood and Built Environment.¹²

KIs were asked how their organization is involved in:

- a. efforts or initiatives to address economic stability issues including poverty, employment, food insecurity and housing instability with their patients and their community.

- b. efforts or initiatives to address areas such as high school graduation rates, enrollment in higher education, language and literacy and early childhood education and development in their community.
- c. efforts or initiatives to address health care and health care access including access to primary care and health literacy in their community.
- d. efforts or initiatives to address social and community context issues such as social cohesion, civic participation, discrimination, and incarceration in their community.
- e. efforts or initiatives to address neighborhood and built environment concerns including access to foods that support healthy eating patterns, quality of housing, crime and violence and environmental conditions in their community.

In all five areas KIs were probed to discuss opportunities by which their organization could contribute more, and they were also asked to identify the main challenges they have met or foresee coming across when addressing these key issues (Appendix C). Prompts were used for each question as necessary to encourage deeper thinking.

Data Analysis Plan

Immediately after each interview, the digitally recorded files were uploaded and saved on a password-protected computer exclusively in the possession of the principal investigator. The interview files were transcribed by transcription software. (Otter.ai) In addition, the researcher captured and documented initial impressions to a Word document after the interview was concluded.

The transcribed interviews were verified against the audio recording to ensure that all thoughts and opinions were included in the analysis. Once verification was complete, the investigator conducted a formal content analysis to identify key themes and categories prior to coding the data. The content analysis used an inductive approach, whereby the information collected in the key informant interviews was used to build broad themes or categories. Following the coding of all interviews, reports were generated for each of the codes to systematically analyze and report on the information received during the KI interviews. Quotes from the KI interviews were used to illuminate or expand on findings or provide different perspectives.

Once the data were analyzed and the study completed, all recordings were destroyed to ensure that no responses would be linked to an individual. The results were presented in the aggregate and the names of the individuals kept confidential. Descriptors of KI's roles are included but not specific roles to maintain confidentiality. Hard copies of data and collateral materials were stored separately in a locked cabinet in the home of the principal investigator.

Delimitations

Only AHCs in North Carolina as defined by the North Carolina Division of Health Service Regulation (DHSR) were included in all phases of this study. The five AHCs in North Carolina as defined by the North Carolina DHSR are UNC Health, Duke University Health System, Vidant Health (now ECU Health), Wake Forest Baptist Health (now Atrium Health Wake Forest Baptist), and Atrium Health (now Advocate Health). Within these health systems, UNC Medical Center, Duke University Hospital, Vidant Medical Center (now ECU Health Medical Center), Wake Forest Baptist Medical Center (now Atrium Health Wake Forest Baptist) and Carolina's

Medical Center are the five academic medical center teaching hospitals in the state of North Carolina.⁸⁵ There are several different frameworks for SDOH. SDOH included in this study were classified as judged to best fit into the Health People 2020 framework. This study focused on how AHCs in North Carolina are looking to contribute to SDOH to improve the overall health of their patients with a goal of improving the health of their communities. This study did not focus on how AHCs are working with defined populations such as part of an insurance plan or ACO although some overlap may exist with findings.

Study Limitations

The author has spent most of his health care career in AHC settings. While this may provide a deeper understanding of some of the opportunities and challenges that AHCs face, this may also introduce biases, related to the contributions AHCs can make to address SDOH to improve the health of their communities. Given that the study is related to addressing SDOH, it is possible that institutions that are less invested in addressing SDOH or building community partnerships were less likely to participate. Similarly, those who have already taken steps to address SDOH or built community partnerships may have been more likely to agree to participate. In addition, interviewing only four stakeholders from each AHCs may not have allowed for the full range of perspectives to be presented and some viewpoints or recommendations may be inadvertently omitted. Data for community benefits was lagging for Form 990 since 2019 forms cover the 2018 calendar year. Hospitals in North Carolina voluntarily submit their community benefit reports to the NCHA. These reports also lag and currently the latest data is for 2019 for all AHCs at the time of writing. The healthcare systems are geographically large and therefore all KIs were asked to focus on the communities of the primary medical centers defined as the county

where the flagship medical center is located for Aims 2 and 3. Only the major medical centers within each system for Aim 1 were studied. The focus on the flagship medical centers rather than the larger health systems is another study limitation. Finally, the researcher's own experiences in addition to career history may bias the results.

IRB Considerations/Confidentiality Issues

The researcher was required to obtain IRB approval from the UNC IRB. As noted above, measures were taken to ensure confidentiality of all research subjects. KI names may have appeared in audio files and written transcripts. These data were kept in password-protected files and were destroyed upon completion of the research. In addition, no names were attributed to the people, but direct quotes were used. Employee identifiers included only worksite location. The researcher does not believe there are psychological risks to employees who participated in the KI Interviews. This risk was further minimized by making participation completely voluntary and making it clear participants may choose not to answer some of the questions.

CHAPTER 4: FINDINGS

Aim 1 Findings

Aim 1: To evaluate the contributions of AHCs in North Carolina to activities benefiting their community.

Community Benefit Reports

A summary of the community benefit reports as reported in the NCHA website for AHCs in North Carolina for 2019 is shown in Table 2.⁸²

The highest dollar amounts of all community benefits are treating charity care, Medicare and Medicaid patients ranging from over \$140 million for Vidant Medical Center to over \$1.5 billion for Atrium Health Carolinas Medical Center. Health professions education ranged between \$48 million for Vidant Medical Center to over \$80 million dollars for Wake Forest Baptist Health. Community building activities were absent for Duke University Hospital (DUH) and UNC, under \$700,000 for Vidant Medical Center, just over \$700,000 for Atrium Health Carolinas Medical Center and a little over \$2 million for Wake Forest Baptist Health.

Table 2 Summary of Community Benefit Reports for AHCs in North Carolina 2019					
	Atrium Health Carolinas Medical Center	Duke University Hospital	Vidant Medical Center	Wake Forest Baptist Medical Center	University of North Carolina Hospitals
Estimated Costs of Treating Charity Care Patients	338,637,343	65,816,004	46,716,641	43,145,095	87,682,002
Estimated unreimbursed costs of treating Medicare patients	852,005,527	185,308,822	93,964,012	141,362,405	51,755,727
Estimated unreimbursed costs of treating Medicaid patients	363,252,171	67,667,988	578,258	19,808,240	0
Estimated unreimbursed costs of treating patients from other means-tested government programs	0	0	0	21,882,350	0
Community health improvement services & community benefit operations	12,165,291	0	3,440,725	6,025,410	0
Health professions education	65,204,339	66,817,681	48,327,095	80,373,558	64,455,817
Subsidized health services	14,731,073	0	0	6,612,850	0
Research costs	23,082,699	0	2,010,157	0	0
Cash and in-kind contributions to community groups	17,192,711	922,362	3,346,885	3,618,837	0
Community Building Activities	722,223	0	662,921	2,134,115	0
Estimated costs of treating bad debt patients **	381,410,948	12,885,223	33,172,742	31,844,098	28,814,862
Total Community Benefits	1,686,993,377	386,532,857	199,046,694	324,962,860	203,893,545

Table 3 lists the itemized benefits as a percentage of the total community benefits for each AHC using the raw data in Table 2. The costs of treating Charity Care Patients, unreimbursed costs of treating Medicare, Medicaid, and other means-tested government programs are by far the largest percentage of total community benefit spending ranging from 70% to over 90%. Health profession education is another leading community benefit spending ranging from 17% to over 30% for all but one AHC in North Carolina. Atrium Health spent nearly 4% of its community benefit total which can be explained by a relatively smaller academic footprint compared to the other four AHCs in North Carolina.

Table 3 - Itemized benefits as % of Total Community Benefit for AHCs in North Carolina 2019					
	Atrium Health Carolinas Medical Center	Duke University Hospital	Vidant Medical Center	Wake Forest Baptist Medical Center	University of North Carolina Hospitals
Community Building Activities as % of Total Community Benefits	0.04%	0%	0%	1%	0%
Community health improvement services & community benefit operations as % of Total Community Benefits,	0.72%	0%	2%	2%	0%
Cash and in-kind contributions to community groups as % of Total Community Benefits	1.02%	0%	2%	1%	0%
Subsidized health services as % of Total Community Benefits	0.87%	0%	0%	2%	0%
Health professions education as % of Total Community Benefits	3.87%	17%	24%	25%	31%
Research costs as % of Total Community Benefits	1.37%	0%	1%	0%	0%
Costs of Treating Charity Care Patients, unreimbursed costs of treating Medicare, Medicaid and other means-tested government programs patients as % of Total Community Benefits	92.11%	82%	71%	70%	67%
Estimated costs of treating bad debt patients as % of Total Community Benefits	22.61%	3%	17%	10%	14%

Table 4 lists the itemized benefits comparing changes between 2018 and 2019 in both dollar and percentage change. UNC did not report certain categories for 2019; when this occurred, these were marked as NA in Table 4. All five AHCs showed an increase in cost for caring for charity care patients ranging from 3% to 53%. There were also increases from 10% to 47% in unreimbursed costs of treating Medicare patients with UNC being the only AHC reporting a decreased cost of 30%. Increases of 2% to 29% for Medicaid patients were reported when data was available. Community health improvement services and operations increased by 14% at Wake Forest Medical Center and by 31% at Atrium Health Carolinas Medical Center; no spending in these areas were listed by DUH and UNC, and Vidant showed a 64% decrease from 2019 spending compared to 2018. Health professions education increased by 29% at Atrium Health Carolinas Medical Center; there were only small increases at DUH (2%), Vidant Medical Center (7%), and UNC (9%) and a small decrease (-1%) at Wake Forest Baptist Medical Center. Subsidized health services increased only at Atrium Health Carolinas Medical Center. Cash and in-kind contributions to community groups decreased at both Vidant Medical Center and Duke Medical Center and had modest increase at Atrium Health Carolinas Medical Center and Wake Forest Baptist Medical Center.

Table 4 - Summary of Community Benefit Reports for AHCs in North Carolina 2019 compared to previous year in \$ and % change										
	Atrium Health Carolinas Medical Center		Duke University Hospital		Vidant Medical Center		Wake Forest Baptist Medical Center		University of North Carolina Hospitals	
	\$	%	\$	%	\$	%	\$	%	\$	%
Estimated Costs of Treating Charity Care Patients	178,262,491	53%	9,058,012	14%	6,030,833	13%	1,355,492	3%	12310820	14
Estimated unreimbursed costs of treating Medicare patients	404,459,369	47%	17,979,762	10%	21,680,926	23%	21,902,848	15%	-15404138	-30%
Estimated unreimbursed costs of treating Medicaid patients	98,674,613	27%	1,210,021	2%	143,650	25%	5,718,101	29%	12739067	NA
Estimated unreimbursed costs of treating patients from other means-tested government programs	NA	NA	NA	NA	NA	NA	**6756669	31%	NA	NA
Community health improvement services & community benefit operations	3,782,047	31%	NA	NA	-2,157,417	-63%	864,640	14%	NA	NA
Health professions education	19,229,264	29%	1,016,663	2%	3,479,082	7%	-859,501	-1%	6074915	9
Subsidized health services	5,428,308	37%	NA	NA	NA	NA	-82,147	-1%	NA	NA
Research costs	190,000	1%	NA	NA	NA	NA	NA	NA	NA	NA
Cash and in-kind contributions to community groups	2,951,000	17%	-458,309	-50%	-756,521	-23%	1,072,162	30%	NA	NA
Community Building Activities	507,475	70%	NA	NA	72,865	2%	1,067,061	50%	NA	NA
Estimated costs of treating bad debt patients	120,547,379	32%	-2,964,579	-23%	7,288,817	22%	-2,191,258	-7%	-1759960	-6
Total Community Benefits	713,484,567	42%	28,806,149	7%	16,328,632	8%	37,795,325	12%	15720663	8
* The ANDI methodology, which uses a facility wide ratio of cost to charges as described in NCHA Community Benefits Guidelines										
** An internal cost-to-charge ratio approach that is based on the methodology specified in the NCHA Community Benefits Guidelines.										

Of note community-building activities increased 70% at Atrium Health Carolinas Medical Center and 50% at Wake Forest Baptist Medical Center but the total dollar values and % as part of total community benefits was very small, 0.04% and 1% respectively (Table 3). Wake Forest Medical Center had the largest increase at over \$1 million while Atrium Health had an increase in spending of just over \$500,000.

The largest increase in spending, in both dollars (\$713 million) and percentage (42%) was by Atrium Health Carolinas Medical Center. UNC Medical Center was a distant second in dollar amount (\$157 million, an 8% increase). Wake Forest Baptist Medical Center spent a distant third in dollars (\$38 million, a 12% increase). Duke Medical Center had an increase in total community benefit spending of nearly \$29 million (7%) while Vidant Medical Center had an increase of just over \$16 million (8%). It is important to remember that community benefits costs remain largely skewed towards treating Charity Care Patients, and unreimbursed costs of treating Medicare, Medicaid, and other means-tested government programs.

Form 990

Form 990 forms for 2018 (most recent forms available filed in 2019) are not available for UNC and Atrium Health Carolinas Medical Center as they are not required to file a Form 990. In addition to their not-for profit hospital status, they are also instrumentalities of states or political subdivisions.

Hospital organizations use Schedule H for Form 990 to provide information on the activities and policies of, and community benefit provided by, its hospital facilities and other non-hospital health care facilities that it operated during the tax year. Form 990 schedule H is only required for organization that answered “Yes” on Form 990, Part IV, line 20a where an organization acknowledges that they operate one or more hospital facilities.⁸⁶ The Form 990 for 2018 for Vidant Medical Center and for Wake Forest Baptist Health did not include a Schedule H on Part IV, line 20a and therefore was unavailable for review. Schedule O explanation for Part III line 4 a provides some insights into their community benefit activities and was used to obtain available information instead.

The only financial information provided for Vidant Medical Center is \$700,000 contributed to the Community Benefits Grants Program. Some examples of community benefits in Schedule O included supporting Healthy Carolinians of the Outer Banks, support for school health partnerships and providing clinical settings for health profession education and support for research activities. Also listed were donations to Meals on Wheels, support for American Red Cross Blood Drives and assisting with fundraising for several organizations including the United Way and the American Heart Association. Some community building activities listed include

supporting road improvements, investing in communication infrastructure through information technology, and supporting the Teen Leadership Academy.

The only specific financial information for Wake Forest Baptist Health lists \$580.7 million to support community benefits. Some specific areas of outreach include the Downtown Health Plaza outpatient medical clinic that offers clinical care and education programs for diabetes and pregnancies for many uninsured and underinsured Forsyth County residents. Other programs listed as innovative ways to reach the underserved include several community-based health clinics sponsored by churches and private organizations, with volunteer assistance from Wake Forest personnel. Faith Health NC is listed as a strong partnership between clinicians, community resources and congregations that seek to avoid readmissions to the emergency department after hospital discharge. The educational mission is also highlighted listing many programs and educational specialties.

Form 990 for 2018 for DUH included a Schedule H explanation. Part I line 7 lists just over \$12 million for cash and in-kind contributions for community benefit (0.37% total expense). The community-building activities Schedule H Part II by specific areas included physical improvements and housing, economic development etc. was not completed. The Form includes information about DUH, Duke Raleigh Hospital and Duke Regional Hospital. This review will focus on the DUH community building activities as the major medical center within the DUH System. DUH helped conduct the community health assessment that identified and develops the health priorities for Durham County in 2017 for 2018 to 2020. The priorities were listed as affordable housing, access to health care and health insurance, poverty, mental health and obesity, diabetes, and food access.

DUH addressed affordable housing by partnering with Habitat for Humanity, the AJ Fletcher Foundation (to develop a downtown area into affordable housing) and the city of Durham to develop an affordable housing trust fund. Increasing access to health care is addressed with multiple programs including Project Access of Durham County (PADC) and Local Access to Coordinated Healthcare (LATCH) which involves a broad partnership of stakeholders to coordinate, and eliminate barriers to, health care. Other programs listed are the Complex Child Program (CCP), the Southern High School Wellness Center and Just for Us (JFU) that provides home care and case management for the elderly and disabled. The DUH System also operates three community health clinics to provide primary care to the underserved.

To address poverty, the minimum wage at DUH was increased to \$14.00 per hour in 2018 and \$15.00 per hour in 2019. The SSI/SSDI Outreach, Access, and Recovery (SOAR) program improves access to disability benefits for those applying for Supplemental Security Income (SSI) and Social Security Disability Insurance (SSDI). The Benefits Enrollment Counseling (BEC) Program assists seniors and those with disabilities and low income to apply for all the benefits for which they are eligible. In 2019 alone BEC assisted nearly 900 individuals for nearly \$6 million in benefits.

Access to mental health and substance use disorder services is another priority where the DUH System partners with multiple community stakeholders. There are several community coalitions including the Durham Crisis Collaborative, the Durham Mobile Crisis Unit, and other partners to provide chronic pain support and standardize opioid clinical use where appropriate. Obesity, diabetes, and food access are addressed and supported through community-based programs such as Bull City Fit. This program includes several community partners and DUH encourages healthy lifestyles for children through multiple activities, education outreach.

Aims 2 and 3 Findings

Aim 2: To explore how AHCs in North Carolina are partnering with local and national organizations to improve the health of patients and their communities.

Aim 3: To identify challenges that AHCs in North Carolina are trying to overcome and enablers to improve the health of patients and their communities.

Aims 2 and 3 were addressed qualitatively through semi-structured key informant (KI) interviews. The KIs were administrators or providers with a practice or research interest in SDOH and population health. Four KI interviews for each of the five AHCs (total N=20). In reporting my findings, I will not identify the specific role or title of the KIs or institution to protect their confidentiality. KI interviews lasted between 33 and 78 minutes with a mean of nearly 49 minutes. KI had between 4 and 38 years of experience with a mean of 19 years. The mean years of experience in their current role was 8 years since several KIs were occupying relatively new roles created by the NC AHCs. The majority of KIs were involved with either strategic decisions involved with community benefit or managing grant dollars related to community benefits. A summary of the KIs roles and titles are summarized in Table 5 below.

KIs directly involved in community benefit decisions largely agreed that decisions were made based on regulatory requirements to provide charity care, as part of their mission to serve as a safety net. Priorities other than charity care and health professions education were based on the findings of the CHNA. AHCs partnered with community benefits departments that had committees or boards to allocate and approve grant funding. Some KIs observed that their organization did not yet have a robust system wide community health strategy across different entities in

the system. Other KIs felt that their organization was pivoting to a more centralized form of decision making for community benefit. Overall, all the NC AHCs were moving towards a strategy with a larger focus on looking for ways to invest further upstream in SDOH motivated by improving health access and health equity.

Table 5 Summary of Key Informants	
Total Number of key Informants	20
Number of Key informants for each AHC	4
Providers at Assistant Professor or above	3
Administrators at Associate Director or Director level	5
Administrators at Associate or Assistant Vice President	4
Administrators at Vice President or President level	4
Other (Program Manager, Vice Chief, Chief, Chief Officer)	5
Title includes "community" or "population health"	11
Title includes "Social" or "Strategy"	6
Title includes "Equity" or "Inclusion"	3
Mean years of experience	19
Mean years of experience in current or related role	8
Involved with strategic decisions on community benefit	10
Involved with managing grants as part of community benefit	7
Not involved with community benefit decisions.	3

Key Findings in Response to the Research Question and Aims

The content of each KI transcript was coded using NVivo. A priori codes were created based on Healthy People 2020 and emergent codes were created during the coding process as described in Chapter 3. The interviews were coded using two cycles to refine emergent codes during the second-cycle coding. Theme saturation was achieved. Probes were used as needed during the KI interviews. Several KIs required little probing and the KIs who did not need prompting gave more coherent answers. Please refer to Appendix E for the Codebook with a priori and emergent themes and descriptions.

Summary of Topics and Sub-Topics

Table 6 summarizes the a priori themes and sub-themes that aligned directly with the Health People 2020 framework that was used in my conceptual model (Figure 1) adapted from the conceptual model of SDOH from Healthy People 2020.

Table 6 – Summary of a priori topics based on Healthy People 2020	
Topics	Sub - topics
Economic Stability	Employment, food insecurity, housing, and poverty
Education	Early childhood and development, enrollment in higher education, high school graduation and language and literacy
Health and Healthcare	Access to health care, access to primary care and health literacy
Social and Community Context	Civic participation, discrimination, incarceration, and social cohesion
Neighborhood and Built Environment	Access to foods that support healthy eating patterns, crime and violence, environmental conditions, and quality of housing

Table 7 summarizes the emergent topics and sub-topics where applicable based on the KI and semi-structured interviews (Appendix C).

Table 7 – Summary of emerging topics based on Key Informant Interviews

Topics	Sub - topics
Effects of Covid Pandemic	Focus on disparities, funding shifts, home health, increased collaboration, telehealth, worsening health status.
Challenges	Benefits cliff, community voice, digital access and technology, lack of trust, multi-sector, politics, reimbursement and payor mix, resources, return on investment, rural areas, segregation and social isolation, silos, sustainability, value-based care.
Enablers	Community partnerships, digital platforms, partnership with local public health, private equity funds and braided funding, SDOH screening, sustained funding, trust and community voice, value-based care.
Education *	Health professions pipeline and primary care, workforce development
Health and Healthcare*	APPs and Allied Health, behavioral health, community health workers, medication access, telehealth and home health, wrap-around services.
Mission	Community benefit, healthcare paradigm shift, patient experience
Neighborhood and Built Environment*	Environmental justice, digital access, transportation
Opportunities	Investing locally, trust building
Regulatory changes	No sub themes
Strategy	Organizational structure, strategic priorities
Teammate resources	No sub themes

* Theme part of a priori themes but sub themes emerged from KI interviews.

Result 1 - Partnerships to improve the health of patients and their communities (related to Aim 2).

KIs nearly unanimously voiced the importance of community partnerships and partnering with local public health to address SDOH to improve the health of patients and their communities. AHCs partner with local public health through Federally Qualified Health Centers (FQHCs), local and state health departments, and various community-based organizations (e.g., food pantries) using community resource hubs or digital platforms that help identify community resources. Food insecurity was the primary SDOH identified and all the AHCs in NC had invested resources over the last several years. Housing was the second most mentioned SDOH that AHCs attempted to address but overall, most KIs agreed they were less able to make a large impact in this area. Most efforts centered on partnerships with Habitat for Humanity and homeless shelters; KIs voiced a desire to do more while acknowledging affordable housing was a statewide and national issue. DUH did invest in some housing decades ago and more recently partnered with the county on a few affordable housing projects. Atrium Health Carolinas Medical Center also invested in an affordable housing through partnerships with the city of Charlotte through the Charlotte Housing Opportunity Investment Fund (CHOIF) Local Initiatives Support Corporation (LISC).⁸⁷ Several other partnerships and close working relationships were identified with other key partners notably school districts, law enforcement agencies and faith-based organizations.

Result 2 - Challenges and enablers for AHCs in NC to improve the health of patients and their communities (related to Aim 3).

Several challenges and enablers were identified by the KIs (Table 7).

Challenges

As the AHCs worked on increasing their wages, the benefits cliff was identified as an issue that was possibly impacting employees with a decrease in total income as they became ineligible for some public benefits due their new incomes. Community voice was identified as crucial to meaningful and successful interventions but often proved difficult to establish. All AHCs were increasing the number of community health workers who, as trusted community members, were seen as an important link between health care services provided and individuals in the community. Digital access was still unavailable in some areas and limited the use of telehealth services and access to digital health records. When digital access was available, comfort with technology still lagged at times. Digital access in this context is defined as internet availability and any modality that supplements in-person care and includes telehealth and patient portals but also includes patient digital literacy. Lack of trust in the community remained a significant challenge and several KIs noted that the Covid pandemic exposed existing health disparities and mistrust of the medical community especially in communities of color. Some SDOH such as affordable housing was a multi-sector problem that was often deemed out of reach for AHCs to tackle alone. Local politics was also a commonly reported challenge as AHCs advocated for change while lack of Medicaid expansion at the state level remained a huge hurdle. Reimbursement and payor mix during the transition from a predominantly fee-for-service approach to value-based care was also a challenge, especially when payors demand quick improvements for problems that require long term investments. Lack of resources included inadequate staffing, education of providers on how to address SDOH, financial resources due to competing priorities. Return on investment was often difficult to demonstrate when addressing SDOH since changes were further

upstream; several KIs discussed the need for new metrics to demonstrate impact for both insurers and the health systems themselves. Rural areas, sometimes within the counties where AHCs were located, had unique challenges including lack of transportation and digital access. Social isolation was also noted by a small number of KIs as another challenge in segregated neighborhoods. Siloed thinking was mentioned several times as a challenge that necessitated a new approach and was often one of the tasks of the KIs. Finally, sustainability was a commonly identified challenge since many interventions were grant funded or under operational streams which resulted in a constant peril for future funding.

Enablers

Community partnerships that were strengthened and expanded during the Covid pandemic were the most identified enabler to address SDOH. Several KIs gave examples of strong partnerships especially when addressing food insecurity in their communities. Digital platforms such as NCCARE360⁸⁸ and FindHelp.org were cited as important tools to link providers with resources available in the community. Partnership with local public health agencies were also strengthened during the Covid vaccination and testing drives; these were also seen as an important way to address SDOH. Collaborations with FQHCs were frequently mentioned as a crucial way to address healthcare for the uninsured. Private equity funds and braided funding were mentioned in the context of addressing affordable housing and seen as an important way AHCs could contribute more to this space. Private equity funding refers to funding from money pooled by investors, whereas when funds are braided two or more funds are coordinated to cover costs and track expenditures. Atrium Health Carolina Medical Center's partnership with the city of

Charlotte was mentioned by some KIs from other AHCs as a model that their AHCs could emulate. SDOH screening mostly through EHRs like Epic was an important first step to identifying social needs. Several KIs mentioned goals to meet a certain threshold for screening but voiced that this is only a first step to linking and providing the necessary assistance to their patients. Sustained funding when available as part of corporate funding, rather than grants, was mentioned as the best way forward for long-term planning. All AHCs had some KIs that pointed to the importance of building trusting relationships with their communities and moving away from thinking that AHCs had all the answers. The move towards value-based care was viewed as a challenge by some, but others saw this shift as an important enabler to addressing SDOH in the long term by payers and the state. Several KIs specifically included Medicaid transformation to a managed care system that specifically addresses non-medical factors including transportation, access to healthy food and stable affordable housing as an important development.

Thematic Findings on SDOH and AHCs in NC

The following are some thematic findings from the KI interviews with pertinent quotes.

Community organization partnerships flourished during Covid.

“We had built a significant partnership anyway with the community prior to COVID. But I think that's really pushed it even more so into almost like a hyper mode of linking.”

“I can tell you the thing to me that was accelerated in COVID that people don't talk enough about, is the collaboration.”

“So, I do see a deeper connection with public health as a result of COVID.”

Community benefit reports do not adequately reflect all AHC activities that benefit communities.

“And then I think, you know, it's a large academic health system. We don't capture it all. You've got entities within a large system like ours that are making contributions and investments in one hospital here may be doing it and ambulatory entity over here may be doing it. That as a system we don't always capture.”

“I don't think we're capturing the monetary value of that as a community benefit, but it certainly does benefit the community. I think it's just on a smaller scale than then would really be worth the trouble of reporting. I suspect that's probably part of it.”

“...because the community benefit report is all about very clean accounting. What I have seen is there's actually a lot of work going on in that space, but it's not always reported in a very neat and clean way.”

“We're looking at adopting a software system.....we've debated about using it for a number of years – it's called CBISA. community benefits inventory for social accountability, about how we're counting community benefits in terms of categories and parameters. That we need to track.”

AHCs in NC are evolving to adapt to healthcare paradigm shifts.

“I truly believe that we're at a point where the health system is trying to shift. It's interesting because the southeast is still predominantly fee for service. And we haven't really moved to value-based care contracts and the way that other areas of the country have and so it's trying to get ahead of that.”

“The accountability is shifting from, you know how many patients you're seeing, to how healthy are those patients within your catchment area.”

“I think I mean; I think it's a big cultural change for healthcare system to think differently about how we spend our dollars.”

“I think sometimes we focus too heavily on ROI for social care and really need to be focusing instead on transforming the model where healthcare has to save money.”

Community voices and building trust are crucial to effectively impact lives.

“And so hearing community voice understanding what type of healing is necessary, really understanding issues from a community perspective, not assuming that we have all of the answers that we are the ones that have all of the solutions.”

“I guess, one piece of the puzzle that I think is important because I think sometimes healthcare institutions just say, hey, we're going to put a clinic right here. We're going to give you what we think you need, and when we think you want, but maybe actually that might not actually align with what the community wants.”

“You know, yes, you remove barriers, but it doesn't matter because they can be a practice sitting in that community that no one goes to. It is about the trust. And so, I just think that that is so important and everything we build out in public health or community health.”

Limited resources and allocation remain a challenge to address SDOH.

“I think certain resources are really hard to get. I mean...like my team community health workers, they want to connect all their patients to the services and resources and support that they can, whether it's from the government, whether it's from organizations, and it's just it's really hard to access certain things.”

“But I don't think there's an understanding of how much would it take to actually make an impact in the community.”

“But we just got to have a way to keep the lights on right.... We can invest in it. But we have to have a way of investing in it: because we don't have unlimited resources either, right we're just trying to break even.”

Grant based funding cannot sustainably fund programs that address SDOH.

“And I think one of the challenges you see a lot of systems come up with is yeah, that's great. But how are you going to fund it and sustain it?”

“And they give out small grants over the course of the year.... I keep saying - you know what I say to them. Are you planting annuals or are you planting perennials?”

“And the way that healthcare especially Community Health is funded, it's usually grant funded so it means everything has a limited time, specific area where you have to be, you can only go to certain audiences that you wrote your proposal.”

“Grants are great, grants are amazing, but if we really want to be able to improve the health of populations over time, we have to get out of this writing a grant and giving money on a very short-term basis.”

Regulatory changes will continue to drive strategies that address SDOH.

“And I think you know, the other thing to it timely that you're working on this right now.... incorporate the JCAHO announcements of screening for social drivers of health and health equity measures, as well as CMS coming out which both I think are looking at January start.”

“Our state is part of Medicaid transformation, RCC (Ratio of Cost to Charges)...but the state was ushering in its Healthy Opportunities Pilot concept (HOP).

“The Medicaid waiver is huge and the whole country is watching that one.”

Summary of Findings

Precise information on how AHCs in NC are contributing to community benefit activities that are closely related to SDOH programs is hard to obtain. The community benefit reports that AHCs voluntarily provide to the North Carolina Healthcare Association website provide valuable insights with total self-reported amounts by category; however, they do not offer details on the types of community benefit activities. Overall, AHCs spend a very small percentage of their total spending on community improvement services and community benefit operations. By far the largest amount spent on community benefit is for treating Charity Care Patients, unreimbursed costs of treating Medicare, Medicaid, and other means-tested government programs. Form 990 schedule H provides more detailed information, but it was only available for DUH. UNC and Atrium Health Carolinas Medical Center as they are not required to file a Form 990, and Vidant Medical Center and Wake Forest Baptist Health did not include a Schedule H on their Form 990 because they did not report operating one or more hospital facilities.

The importance of community partnerships and partnering with local public health to address SDOH to improve the health of patients and their communities cannot be overstated. These partnerships were strengthened and flourished with the Covid pandemic. Food insecurity and access to health care were the most addressed SDOH. Housing was viewed as a crisis that needed to be addressed on a statewide and national level. Several challenges to addressing SDOH were identified including connecting with and building community trust, digital access, reimbursement, payor mix, and sustaining programs and interventions. Enablers included strong community partnerships, digital platforms connecting patients to services, private equity, and

braided funding. The transition to value-based care remains both a challenge and an opportunity for NC AHCs to address SDOH.

CHAPTER 5: DISCUSSION OF STUDY FINDINGS

Funding sources for initiatives to address SDOH and establish meaningful community partnerships are two key emerging areas of focus for NC AHCs as they seek to improve the health of their communities. Adapting to regulatory changes, especially Medicaid transformation and the shift towards value-based care, are critical to achieve this goal.

Funding Sources

Funding for initiatives to address SDOH remains a challenge for NC AHCs. Moreover, community benefit reports fail to capture the monetary value for all activities in which AHCs engage to benefit communities. KIs in this study had concerns that the community benefit reports may be missing important efforts and contributions that AHCs are making towards improving the health of their communities. However, based on the available community benefit reports, AHCs are spending a very small percentage of their total spending on community improvement services and community benefit operations. The largest amounts of reported community benefit dollars are for treating Charity Care Patients, unreimbursed costs of treating Medicare, Medicaid, and other means-tested government programs.

Limited resources and allocation remain a challenge to address SDOH. Grants are often the source of funding for innovative initiatives addressing SDOH; however, grants cannot sustainably fund these programs. AHCs frequently address SDOH with short-term funding especially when isolated initiatives in different pockets of an organization submit grants by individual or small groups of clinicians with specific interests. KIs were very grateful that grant dollars supported the start of new initiatives but had concerns about sustainability long term. Some KIs voiced concerns when programs are funded from the operational budget rather than a portion of long-term organizational dollars for specific purposes.

KIs also discussed the importance of braided funding with private equity investment. This was noted to be particularly important when AHCs participate in broader initiatives such as seeking solutions to increasing affordable housing. Several KIs discussed partnering with Habitat for Humanity, but they also realized these efforts have minimal long-term impact on the problem. KIs from different AHCs cited efforts by Atrium Health Carolinas Medical center as an example of a large-scale initiative when, in 2019, they partnered with LISC through the CHOIF as a joint effort among Foundation for The Carolinas, the City of Charlotte, corporations, and philanthropic organizations to tackle the city's supply and quality of affordable housing.⁸⁷

Community Partnerships

The majority of KIs shared that SDOH screening initiatives through their EHR resulted in better SDOH data for their organizations. However, several KIs also voiced a concern that connecting patients with identified social needs to the appropriate community organization or resources was challenging. Platforms with linkage to community partners were helpful, with

NCCARE360 and FindHelp.org often cited. However, communicating bi-directionally with small non-profit organizations or local public health departments still proved challenging due to digital limitations for many small organizations.

Several KIs emphasized that hearing community voices early on and building trust are crucial to developing programs that incorporate SDOH. Communities need to lead gap analysis facilitated by the AHCs to link with the necessary resources in the community. Key community members' involvement must begin early and continue through implementation and program impact evaluation.

Community organization partnerships flourished during Covid starting with organizations that provided Personal Protective Equipment (PPE) and strengthened with testing and vaccination drives. Successful vaccination drives depended heavily on linking the resources of the AHCs with the trusted voices in the community. Previous partnerships were strengthened with public health entities partially because Covid brought the intersection of public health and healthcare systems to the forefront. New partnerships were also created as food insecurity, homelessness and poverty were exacerbated with the general economic downturn associated with the pandemic. The pandemic created opportunities for new partnerships, but whether those will grow or be sustained is uncertain.

Regulatory Agencies

Regulatory changes will continue to drive the pace and direction for strategies and initiatives that address SDOH. The Joint Commission created a new standard in the leadership chapter that has six new elements of performance (EP). The intent is to highlight health disparities as a

patient safety and quality issue. The second element (EP2) specifically requires organizations to assess health-related social needs and provide information about available community resources. The SDOH to be assessed are not specified but examples include food insecurity, housing insecurity, and transportation access. This requirement became effective on January 1, 2023.⁸⁹ CMS reaffirmed its commitment to expanding the collection, reporting and analysis of standardized data including SDOH. Connecting individuals with social services and collecting and leveraging data for underserved communities is one of five priorities to increase health equity for all individuals as part of its Framework for Health Equity for 2022-2032.⁹⁰

Several KIs also highlighted Medicaid transformation initiatives as an important space where changes are occurring and will continue to shape the pace and direction of changes in the SDOH space. Most Medicaid services in NC formally moved to NC Medicaid Managed care on July 1, 2021.⁹¹ The Healthy Opportunities initiative specifically addresses SDOH as part of the DHHS mission “of improving the health, safety and well-being of all North Carolinians.”⁹² The areas of focus are housing stability, food insecurity, transportation access and interpersonal safety. Healthy Opportunities include several strategies, including developing a standardized list of screening questions, building a structure to develop and grow the use of community health workers, and developing a statewide map of SDOH indicators to guide priorities.⁷⁹

NCCARE360, also part of this initiative, was mentioned by several KIs as an important resource to connect patients with identified SDOH needs to community resources.⁸⁸ Another initiative mentioned by several KIs was the Healthy Opportunities Pilot, a program that tests and evaluates the impact of interventions directed at SDOH. The pilot areas currently do not include the counties where most NC AHCs are located other than Pitt County where ECU Health Medical Center (formerly Vidant Medical Center) is located. However, this is an important evaluation that all the

AHCs are watching closely since NCDHHS has indicated it would systematically expand the program across the state if shown to improve outcomes and reduce costs.⁷⁹ Finally some KIs lamented that the political battle over Medicaid expansion is another important issue that they are watching carefully. NCDHHS reports that expanding Medicaid in NC would give access to more than 600,000 uninsured North Carolinians (especially in rural areas) and would save lives. Of particular importance to NC AHCs with large expenditures on charity care, NCDHHS also cites a boost to uncompensated costs to health care providers who care for uninsured people.⁸¹ NC remains one of only 11 states that has not yet expanded Medicaid. In June of 2022 the NC Senate passed a bill that would have expanded Medicaid, but the House never voted on that bill.⁹³

Health Care Paradigm Shift

AHCs in NC are evolving to adapt to healthcare paradigm shifts but overall KIs felt that NC and the broader southeastern US is still in primarily in a FFS system while working on continuing to grow value-based contracts where applicable. Progress was being made with data collection for SDOH and strengthening of community partnerships. Many grant-funded initiatives were led by small numbers of researchers and clinicians interested in SDOH. Traditionally, initiatives were led by departments of community health, population health, faith health and health equity. More recently some AHCs, like Atrium Health and ECU Health (formerly Vidant Health) created high administrative positions directly related to social impact strategy that directly addresses SDOH and social needs. The majority of KIs believed that AHCs could and should do more to address social needs and SDOH but cite the barriers discussed, including lack of resources and misalignment of financial incentives.

SDOH Key Areas

KIs listed several examples of initiatives by NC AHCs addressing SDOH. Table 8 summarizes the level of activity, and examples of initiatives surrounding SDOH sub-topics as defined in Health People 2020.

Table 8 – Summary of SDOH Initiatives		
SDOH Sub-topics	Initiatives surrounding SDOH sub-topic	Examples of initiatives
Employment	Moderate	Living wage for employees, Opportunities for advancement, partnering with community colleges
Food Insecurity	High	Partnering with food banks, food boxers for patients and employees, farmer's markets, summer lunch programs for children, food rescue programs, food prescription programs.
Housing	Moderate	Affordable housing for employees, partnering with Habitat for Humanity, braided funding for affordable housing, attempting to reduce evictions.
Poverty	Minimal	Charity care to assist the uninsured, identifying neighborhoods with high social vulnerability.
Civic Participation	Minimal	Encouraging volunteer participation from employees
Discrimination	Moderate	Training for employees in Diversity, equity and inclusion, improving diversity in hiring and promotion practices, advocacy efforts.
Incarceration	Minimal	Employment of previously incarcerated individuals, Assistance for recently released individuals.
Social cohesion	Minimal	Partnering with neighborhood organizations.
Access to foods that support healthy eating patterns	Minimal	Working with farmers in food deserts, education for how to access healthy food through CHWs.
Crime and violence	Minimal	Violence intervention programs.
Environmental conditions	Minimal	Partnering with EPA for water quality, Looking at sustainable practices at the medical centers.
Quality of housing	Minimal	Working with asthma patients for housing quality, partnering with CBO to improve housing quality.
Early childhood and development	Minimal	Early childhood education, English as a second language.
Enrollment in higher education	None	Largely seen as role of the universities.
High school graduation	None	Largely seen as role of the universities.
Language and literacy	Minimal	Working with CHWs to decrease language barriers.
Access to health care	High	Mobile clinics, pop-ups, telehealth, increase in CHWs, partnering with FQHCs, financial assistance for patients.
Access to primary care	High	Brick and mortar primary care clinics, increased number of APPs,
Health literacy	Minimal	Education through CHWs.

Addressing access to health care and expanding primary care was universally viewed as a direct responsibility of AHCs. All the KIs cited expanding coverage through brick-and-mortar expansion and more recently through telehealth and taking health care directly to communities through pop ups, mobile clinics, and other initiatives. All the KIs also cited the importance of their contributions to economic stability. AHCs are one of, if not the, largest employers in their communities, and therefore directly contribute to employment opportunities and improving pov-

erty. Food insecurity initiatives were by far the most often cited and could be addressed by partnerships with local food pantries, food boxes for employees and patients, programs to provide meals for children during the summer months, and a food prescription program.

Initiatives addressing education were often viewed as mostly an area best served by the universities and not AHCs. A few KIs were aware of some collaborations around language and literacy, and early childhood education and development. None of the KIs could cite initiatives intended towards improving high school graduation rates. AHCs supported education advancement for their employees through tuition reimbursement and incentives to develop professionally.

Several KIs saw the role of AHCs as more of a facilitator with content expertise and resources rather than directly addressing broad areas of social and community context, neighborhood, and built environments. KIs had little knowledge of activities surrounding civic participation and social cohesion. UNC's Formerly Incarcerated Transition Program (NC FIT) was cited as an example that has expanded to Durham County and Mecklenburg County where DUH and Atrium Health Carolina Medical Centers are located respectively. The program employs community health workers who have a history of incarceration and were recently released to assist recently released individuals with health care needs.⁹⁴ Discrimination was mostly addressed with employees through required diversity and equity initiatives, but not at a broad community level. Several KIs voiced a heightened awareness by their organizations of the role of structural racism in health care that was heightened with the Covid pandemic. In a recent viewpoint paper in the New England Journal of Medicine, Nundy et al argue that with the Covid pandemic and national attention on systemic racism, health equity should be a fifth aim for quality health improvement.⁹⁵ Adding health equity to the current quadruple aim would alter how we address quality

improvement activities and initiatives. This approach was echoed in a recent National Academy of Medicine report in which the authors proposed an equity paradigm shift that intentionally asks all healthcare stakeholders, including AHCs, to use technology to advance equitable health outcomes.⁹⁶

Crime and gun violence was another area where KIs had a strong interest in developing interventions but had limited knowledge of any current large initiatives. Many KIs cited treating victims of gun violence through their emergency rooms since they all had Level 1 trauma centers. Several KIs cited partnering with local law enforcement or local government officials to see how they could become part of a broader solution.

Housing was another area where AHCs had mostly participated in local partnerships with Habitat for Humanity, but KIs realized that more needs to be done. The debate of how involved AHCs should be with issues like housing instability and homelessness continues. Most KIs acknowledged AHCs need to become more involved in larger scale solutions, but these are complex problems that AHCs cannot tackle alone. Some KIs cited Atrium Health Medical Center's initiative to expand affordable housing as the direction AHCs may need to follow.⁸⁷

Recommendations for NC AHCs

AHCs are continuing to build and strengthen their community partnerships. This is occurring in the context of changing state regulations and a shift towards value-based care. Based on these findings my recommendations for AHC are:

1. Screen for SDOH and connect patients with identified social needs to the appropriate community organization or resources with platforms with linkage to community partners.

2. Hear community voices early on when launching new initiatives to build trust.
3. Strengthen previous partnerships and build new ones with school districts, law enforcement agencies and faith-based organizations.
4. Increase the number of community health workers who, as trusted community members serve as an important link between health care services provided and individuals in the community.
5. Follow regulatory changes to anticipate the direction for strategies and initiatives that address SDOH.
6. Advocate and invest in improving digital access for the community.
7. Provide living minimum wage and affordable housing for employees to staff key areas.
8. Invest resources in new metrics to demonstrate impact and return on investment for both insurers and the health systems themselves.
9. Explore braided funding with private equity investment - particularly important when AHCs participate in broader initiatives such as affordable housing.
10. Advocate for Medicaid expansion and expanded roles for APPs.
11. Restructure the organization to decrease siloed thinking and prioritize social impact strategies.
12. Fund initiatives and programs addressing SDOH through budgets not linked to operational dollars.
13. Add social accountability to core mission statements and vision.

Limitations of the Study

There are biases in the quantitative data. Community benefits reports for NC AHCs are voluntarily submitted to the NCHA and lack important details. Limited information was obtained from Form 990. Form 990 forms for 2018 (most recent forms available filed in 2019) are not available for UNC and Atrium Carolinas Medical Center as they are not required to file them. The Form 990 for 2018 for Vidant Medical Center and for Wake Forest Baptist Health did not include a Schedule H and therefore only DUH had a Schedule H available to review. The qualitative findings informed the quantitative results. The qualitative findings indicated there were several ongoing initiatives and suggested there may be more investment than the reported quantitative data. The quantity and breadth of initiatives discussed by the KIs during the interviews is far more recent than the latest community benefit reports that were all completed pre-pandemic.

Interviewing only four stakeholders from each AHCs may not have allowed for the full range of perspectives to be presented. Some viewpoints or recommendations may be inadvertently missed because it is hard for each KI to be aware of initiatives across the wide range of SDOH defined by Health People 2020. KIs reported a large amount of activity around some areas such as food insecurity; however, it is hard to quantify the initiatives since this was dependent on a small number of KIs for each organization. The healthcare systems are geographically large and therefore all KIs were asked to focus on the communities of the primary medical centers defined as the county where the flagship medical center is located. The focus on the flagship medical centers rather than the larger health systems is another study limitation. Finally, the researcher's own experiences may bias the results.

Recommendations for Further Research

NC AHCs have very broad, scattered initiatives that often lack a plan for impact and evaluation. That is the primary area where further research is required. Rigorously modeling return on investment is also important but hard to measure in the near term. Return on investment needs to be tied to value added to the delivery of care which cannot be measured in only financial terms. In depth case studies for individual AHC initiatives on a system level including organization structure to implement initiatives is another area for further research.

CHAPTER 6: PLAN FOR CHANGE

The results of this study will inform a plan for change for NC AHCs to identify near- and long-term goals for improving the health of the community by targeting SDOH. The plan for change presented here can serve as a blueprint for a NC AHC embracing the goal of incorporating SDOH to improving the health of their communities and making this goal part of its integral core mission. To that end, I will utilize Kotter's 8-step process for leading change (Figure 2) on how to move the needle for organizations that have not taken any steps in this direction or those that need to reinvigorate their efforts and accelerate change.⁹⁷



Figure 2 The 8-Step Process for Leading Change - Kotter

Making population health initiatives around SDOH an integral part of the core mission of an AHC would require creating a new strategic vision and initiatives consistent with its financial health. Accordingly, my plan for change will focus on creating a sense of urgency, building a guiding coalition, forming a strategic vision and initiatives, enlisting a volunteer army, enabling action by removing identified barriers, generating short-term wins, sustaining acceleration, and ultimately instituting change using the Kotter model. The long-term goal would be to establish and sustain partnerships and active programs addressing SDOH to improve the health of the community, with a focus on community-building activities as part of a social accountability mission. A summary of the plan for change is presented in Table 8, highlighting recommended actions for each step in the Kotter process and suggested measures or indicators of success.

Create a Sense of Urgency

The Covid-19 pandemic highlighted existing health inequities; and other coinciding events like the George Floyd and Brianna Taylor killings further brought national attention to the larger issue of systemic and structural racism. Connecting SDOH to health outcomes has become more part of the social discourse, but it is imperative that we move beyond merely collecting data on SDOH. Instead, SDOH data must be used to identify SDOH as a quality and safety issue that, when addressed, can improve health outcomes, narrow health disparities, and promote health care equity. As NC AHCs continue to collect SDOH data, they should publicize their data and highlight current areas where major improvements are necessary. As more data on food insecurity and housing instability are collected, sharing both aggregate results and individual stories will create a sense of urgency to address SDOH. To monitor progress and momentum for this especially important first step for change, NC AHCs should track presentations and communications with internal and external stakeholders to create a sense of urgency and prepare for the subsequent steps necessary for change. Surveying key executives and senior administrators on issues surrounding SDOH can also add to the sense of urgency.

Build a Guiding Coalition

Building a guiding coalition requires the executive team to reach out and partner with a broad array of external stakeholders including public health, local and state government departments, not for profit organizations and business leaders. Before doing so, the executive team will need to identify key stakeholders within their organization's departments to lead effective change across the organization. These individuals are critical to building coalitions with community

partners, local government, local businesses etc. AHCs have traditionally not always fully leveraged the expertise and resources of their academic departments in public health, social work, nursing etc. Blending the academic and research expertise should be a key part of an AHC guiding coalition strategy. As the momentum for building this guiding coalition progresses, we expect increased active participation of AHC administrators on committees, boards etc. and sustained academic/clinical/community partner working groups.

Form Strategic Vision and Initiatives

As part of my plan for change, NC AHCs will need to add social accountability as part of their core mission in addition to the traditional missions of clinical care, research, and education. Strategic initiatives can then be aligned with a new lens of social impact to plan for a future where AHCs will address SDOH. A new strategic vision will lay the groundwork to motivate members of the guiding coalition and align all initiatives with addressing SDOH. Organizational restructuring or realignment may be necessary to make social accountability a priority at all levels. Organizational core values must be revised and branded to incorporate social impact and accountability as part of the larger vision and strategy. Communicating the new vision and strategy with all internal and external stakeholders, in a clear simple message will tie actions and initiatives to purpose. Visuals and graphics can help define and reinforce strategic vision. Successfully engaging the workforce can unite an entire health system as employees rally behind a common cause.

Enlist a Volunteer Army

As AHCs successfully rally employees behind a new strategic vision, a growing volunteer army that is empowered to act can build the necessary momentum to create change. Clinicians interested in SDOH can partner with employees in their departments and with academic researchers and administrators in population health and other departments including nursing, public health, and social work. Together the volunteer army can implement the message from senior leadership across the organization to reach a critical mass. Presenting and facilitating more opportunities for employees to participate in initiatives will increase the likelihood of success. Recognizing and celebrating efforts and new initiatives will be important to keep employees engaged and increase participation in initiatives. Tracking and following the engagement of employees and level of participation can demonstrate a growing momentum.

Enable Action by Removing Barriers

Identifying and removing barriers through engaging stakeholders is the key to enabling action, removing silos, and launching initiatives successfully. Bureaucratic barriers and inefficient processes within the AHCs can stifle change. Barriers identified in this study that limit partnerships with external stakeholders include failing to listen to community voices, build trust, and fund initiatives in a sustainable manner. Organizational leaders must engage their employees to identify current barriers that have impeded past initiatives. Processes may need to be improved to empower teammates to work across departments and policies may need to be implemented to enhance relationships across stakeholders. As barriers are diminished or removed, new workflow

processes and collaboration across departments and stakeholders will position AHCs to address SDOH in a meaningful manner that improves the health of their communities.

Generate Short-Term Wins

Generating and communicating short-term wins is another important step to drive and accelerate change. Early in the process, NC AHCs must communicate their success with meeting SDOH screening targets, but more importantly they should publicize success stories across organizations to further engage their volunteer army and external stakeholders. Tracking new actions taken, new processes, lessons learned, success stories, and progress towards short-, intermediate-, and longer-term goals for new initiatives can also show tangible progress. Dashboards with progress towards goals can quantify progress but more importantly the success stories of improved outcomes and better health for parts of the community can energize volunteers and further engage external stakeholders. Sharing results and success stories through internal communications and public media can increase the impact of successful initiatives.

Sustain Acceleration

Sustaining acceleration once change is initiated can be a challenge, and Kotter warns that “pushing complacency a second time can be much more difficult than it was the first”⁹⁷ Sustaining acceleration largely includes continuing the previous steps in the change process including maintaining a growing the volunteer army, continuing to eliminate barriers, and to generating and publicizing wins. As NC AHCs pivot more towards value-based care, creating sustainable

funding sources and organizational structures will need to be revised, particularly if health systems grow. Wins with successful initiatives addressing SDOH must be used to increase momentum to further realize the new strategic mission. At this stage, successful change will produce teams that demonstrate the ability to stay the course after the initial launch of programs. Continuous impact evaluation of initiatives should demonstrate continuous improvement towards stated goals.

Institute Change

Incorporating social accountability at all levels of an AHC is a transformational change and will inevitably be a long process. Regularly communicating new behaviors and showing how they are connected to better performance will be key to instituting change. As the previous traditional structures adapt to the new strategic vision, a new organizational structure incorporating social accountability at all levels will be necessary. AHCs are large complex organizations and the transition towards social accountability will require leveraging all aspects of the health system towards change. As change is instituted, new processes arising from the new strategic vision can create a new network of community partnerships, improving overall quality of care.

Table 9 Summary of Plan for Change

#	Step	Recommended Actions	Results/Measures of Success
1	Create a sense of urgency	Connect SDOH to health outcomes. Promote addressing SDOH as a quality and safety issue that increases health care equity. Publicize current areas where improvement is necessary e.g., food insecurity, housing instability etc.	Track presentations, publications for promotion of addressing SDOH with set goals. Pre and post knowledge test for key groups e.g., senior administrators.
2	Build a guiding coalition	Identify individuals in various functional departments to lead effective change. Leverage university academic departments in public health, social work, nursing etc. Build coalitions with community partners, local government, local businesses etc.	Active participation of AHC administrators on committees, boards etc. Creation of and activity of academic/clinical/community partner coalitions.
3	Form a strategic vision and initiatives	Revise organizational core values and incorporate social impact in a vision and strategy. Communicate new vision and strategy at all levels with all stakeholders.	Communicate new vision in simple message. (Graphics, mission statement etc.) Tie actions and initiatives to communicated strategic vision.
4	Enlist a volunteer army	Cascade message from senior leadership to all different levels to reach employees. Empower employees at all levels to spread and act on new vision. Recognize and celebrate efforts and new initiatives.	Increased excitement for employees to be part of initiatives. Track engagement of employees, demonstrating an increased level of participation.
5	Enable action by removing barriers	Identify current barriers that have impeded past initiatives. Improve processes to empower teammates to work across silos and departments. Adapt or change organizational rules to increase access to all stakeholders.	Barriers identified and solutions created through evaluation. New initiatives across different stakeholders previously siloed. New workflow processes and linkage across departments.

6	Generate short term wins	Track actions taken, new processes, lessons learned, success stories across organizations. Include short, intermediate, and longer-term goals for new initiatives across organizations.	Dashboards with progress towards short term goals. Sharing of results and success stories through internal communications and communications with stakeholders.
7	Sustain acceleration	Celebrate wins, publicize plans, and use momentum to further realization of strategic mission. Maintain and increase support for all new processes and initiatives long term. Enlist new volunteers and revisit sense of urgency activities if necessary.	Teams demonstrate the ability to stay the course after the initial launch of programs. Evaluation of initiatives demonstrates continuous improvement towards goals.
8	Institute change	Communicate new behaviors and how they are connected to overall change, and better performance. Synchronize previous traditional structures with the new strategic vision. Leverage all aspects of the health system towards change.	New organizational structure incorporating social accountability at all levels. New processes and a new network of community partnerships, improving overall quality of care.

Personal Actions to Drive Change

In my current leadership role as the Director of Nurse Anesthesia at Atrium Wealth Wake Forest Baptist in Winston Salem NC, I will take several steps to make this change happen at my organization.

To create a sense of urgency I will share my findings with my leaders at Wake Forest and Atrium Health to identify immediate action items that align with institutional priorities. As part of building a guiding coalition I will connect with my Advanced Practice Provider (APP) colleagues and my new connections with KIs to identify and promote promising practices and grow

a coalition at Wake Forest. I will work with the nurse anesthesia faculty to use their connections with Wake Forest University to incorporate social accountability and communicate this new vision as part of forming a strategic vision. I will also enlist a volunteer army of other APPs, faculty interested in SDOH and my new connections with KIs to build a network that incorporates SDOH in their practice. I will also explore how we can remove barriers and improve resources to address SDOH with our patients at the pre-anesthesia clinic and work with the APPs in the pre-anesthesia clinic and other primary care clinics to generate and share success stories. To sustain acceleration, I will continue to produce and distribute information by presenting at lectures and anesthesia grand rounds and supporting others' work. I also intend to publish my dissertation research to disseminate my findings. Finally, to institute change, I will focus on my sphere of influence at the pre-anesthesia clinic and my APP colleagues at Wake Forest and Atrium Health while also exploring other opportunities to institute change at an organizational level.

Conclusion

The plan for change will ultimately depend on how successful we are at considering social care as part of health care. The National Academies of Sciences, Engineering, and Medicine outlines five goals for healthcare systems integrating SDOH into health care with recommendations for each goal. The first goal is to design health care delivery differently to integrate social care through awareness, adjustment, assistance, alignment, and advocacy. The other goals are building a workforce specifically integrating social care, developing an interoperable digital infrastructure between social care organizations and the health system, financing social care as part of health care and funding and conducting impact evaluation of social care initiatives.⁹⁸ AHCs

can use these recommendations and others when creating a new strategic vision incorporating social accountability at all levels. AHCs face a steep challenge to incorporate social accountability at a core mission level. Maintaining adequate margins to fund their other missions of clinical care, research, and education, while pivoting from a fragile FFS system to a value-based system will require planning, innovation, and perseverance if we want to achieve improved health outcomes.

APPENDIX A: TABLE 1

Table 1 - Academic Health Centers and Social Determinants of Health: Community Health Improvement

Article ID	Author (Year)	Study Design	Study population and setting	Sample size	Methods/data collection, analysis	Intervention	Outcomes Measured	Results	Limitations	Author conclusions	Quality
1	Aguilar-Gaxiola et al. 2014	NA Taxonomy proposal	The Clinical and Translational Science Awards (CTSA) Consortium's Community Engagement Key Function Committee	NA	A unified approach to monitor progress in improving population health outcomes.	Development of a taxonomy of community health indicators.	NA	The proposed taxonomy goes from broadest to more specific at three levels 1) determinants of health, 2) community health indicators and 3) specific quantifiable measures.	Lack of integration since currently there are multiple community health indicators which are at times conflicting.	The proposed taxonomy is meant to develop common metrics for CTSA's and community partners to better measure changes in population health.	Low - Taxonomy proposal
2	Alberti et al 2018	NA Opinion piece	NA	NA	NA	NA	NA	NA	NA	Ethical obligation to address issue of social and economic injustice because they threaten the health of the communities that AHCs serve.	NA - Opinion piece
3	Battaglia et al. 2020	Cluster-randomized, stepped-wedge hybrid type I effectiveness-implementation study design	Boston residents at risk for delays in breast cancer care in four AHCs in Massachusetts.	Expectation of expect approximately 1100 total patients over the full intervention period. With approximately 500 historical control patients, and 550 intervention patient	Translating Research into Practice (TRIP) utilizing community engaged research methods to address breast cancer care delivery disparities. Primary clinical outcome abstracted from EHR. Mixed methods to collect the secondary implementation outcomes.	Three evidence-based strategies: patient navigation services, a shared patient registry for use across academic medical centers, and a web-based social determinants of health platform to identify and address barriers to care.	Time to first treatment and receipt of guideline-concordant treatment.	NA. Implementation is ongoing over a 5 year period.	Not addressed	The goal of this study is to provide a model of how community-academic partnerships can drive innovations in information sharing and systems implementation to address health disparities.	Moderate - cluster randomized implementation study. May not be generalizable and ongoing.
4	Bedoya et al. 2018	Evaluation of Interprofessional Student Hotspotting Learning Collaborative qualitative design.	Virginia Commonwealth University students who participated in the Interprofessional Student Hotspotting Learning Collaborative to train students in working as an interprofessional team (Hotspotting)	20 reflections from 5 students from the VCU program working with 5 complex patients.	Across 10 sites, 57 students participated during June-December 2014 with 244 reflections across all sites. The article focused on the VCU experienced	The Collaborative trained health professions students to succeed in the Hotspotting approach	Benefits of Hotspotting approach.	The review of the VCU students' experience revealed that the Hotspotting program was successful. Learning about social determinants affect health and the benefits of interprofessional teamwork was an effective way to address the unmet health and social needs of complex patients.	The study only evaluated the student perspectives and further study evaluating the impacts of the program on patients and the community may be necessary.	The VCU experience demonstrated that the Hotspotting program was successful. Deficiencies of the program's initial curriculum were also identified and ideas for enhancing the program structure were also obtained from the student feedback.	Low - small sample size
5	Berkowitz et al. 2016	NA Case study	Johns Hopkins Community Partnership (J-CHP)	NA	NA	NA	Community based intervention, an acute care intervention bundle and a skilled nursing facility intervention.	The AHC partnered with two Community based organizations (CBOs) and five skilled nursing facilities to facilitate transitions to post-acute care.	NA	Authors stress the need for AHCs to continue to innovate to meet the needs to their patients and their communities.	Low - case study
6	Derham et al. 2013	Retrospective cohort study	Carolina Health Net a partnership between University of North Carolina (UNC) Department of Family Medicine; UNC Health Care System (UNC HCS), and Piedmont Health Services (PHS), a CHC with six practices in a four-county region surrounding UNC HCS.	A cohort of 1,958 CHN patients enrolled as of June 30, 2012 was analyzed to evaluate the program. A second cohort of 1,788 patients who had been enrolled in CHN for greater than one year as of June 30, 2012 was also examined.	Analysis of billing data	Development of a partnership between an AHC and Community Health Center - Providing medical homes, high-risk case management, transitioning patients discharged from ED and inpatient hospital settings, coordinating subspecialty care and assisting the uninsured with Medicaid applications	Decrease cost of care and utilization.	4,400 uninsured patients connected to primary care services, 2,200 uninsured patients have been assisted to enroll in Medicaid. ED visits declined by 47% and ED charges decreased by 41% across the cohort of CHN patients after CHN enrollment. For the cohort enrolled of longer than one year patients hospital admission rates were 31% lower, hospital days were 38% lower and hospital charges were 23% lower. CHN has a fiscal return on investment for UNC HCS of approximately \$750,000 per year.	Incompatibility of information systems, comparing demands and pressure to demonstrate an early return on investment.	The authors felt this program was successful and This project can serve as a model for other AHCs looking to partner with community-based providers to improve care and control costs for underserved populations.	Moderate - Large sample size. Results may not be generalizable.

7	DeVoe et al. 2017	NA Opinion piece	NA	NA	NA	NA	NA	NA	NA	Authors outline steps for AHCs to move towards promoting health by creating an Institute for Healthy People based on the pillars of AHCs, community laboratories and community classrooms.	Low - Opinion piece
8	Fagnan 2017	NA Case study	The Health Extension Regional Officers (HEROs) program at the University of New Mexico	NA	NA	NA	NA	The Oregon Rural Practice Research Network uses four pillars to engage the community: community engagement, practice transformation, research, and education.	NA	The health extension model offers many potential benefits, including increased engagement of communities and stakeholders at many levels, the development of research partnerships between academic health centers, practices, and community agencies, and the integration of multiple efforts to ultimately improve the quality of care.	Low - Case study
9	Fiori et al. 2019	Pilot program implementation using implementation science framework.	Pediatric clinic at a federally qualified Community Health Center in Bronx County New York	N=7266 (active CHCC pediatric patients during the study period from May 1, 2017 through April 1, 2018)	Process mapping, development of new workflows and performance improvement methods. EMR for screening data and ICD-10 codes for eligible visits.	Community Linkage to Care (CLC) that integrates social needs screening and referral support using community health workers (CHWs) as part of routine primary care visits.	SDH screening and adoption rate by providers. (78% of providers had their patients screened on more than half of eligible well-child visits)	A total of 4162 SDH screens were conducted at 6410 eligible well-child visits for an overall screening rate of 65%. 19.7% (820/4162) of the SDH screens performed had one or more positive responses. childcare 48.8% (400/820), housing quality and/or availability 39.9% (327/820), and food 22.8% (187/820).	Limited generalizability since site already had buy in for impact of social needs, and grant funding for CHWs. Metrics extracted from EMR may have missed screening data.	Screening for social needs at well-child visits is feasible as part of routine primary care.	Moderate- Implementation study with large sample size but limited generalizability.
10	Fos et al. 2019	Cross-sectional study using secondary analysis of published community benefit reports.	North Carolina's tax-exempt hospitals.	Fifty-three private, nonprofit hospitals.	Community benefit was categorized using the North Carolina Hospital Association's community benefit reporting guidelines. Multiple regression analysis using generalized linear model was used to examine the variation in community benefit spending among study hospitals considering differences in hospital-level and community characteristics.	Financial analysis of reports	Dollar expenditures as a percentage of operating expenses of the 2 categories of community benefit spending: patient care financial assistance and community health programs.	Community benefit spending was \$2.6 billion, 85% for patient care financial assistance, and only 0.7% of total spending allocated to community-building activities. Community benefit spending was equivalent to 14.6% of operating expenses. Hospitals with 300 or more beds provided significantly higher investments in community health programs (% of operating expenses) than smaller hospitals. Access to care was not associated with patient care financial assistance (p = .81) or community health programs expenditures (p = .9).	Most of the community benefit spending was on patient care financial assistance (individual welfare) rather than population health improvement initiatives. Very little investment in community-building activities that address SDOH is occurring.	Moderate - cross sectional study design analyzing community benefit reports using multiple regression analysis.	
11	Georgiev et al. 2019	NA - implementation of a new community paramedicine transitional health support (CP-THS) program.	NA - implementation of a new community paramedicine transitional health support (CP-THS) program. Setting is urban academic medical center	N= 230 patients	Implementation of new paramedicine program.	Eligibility criteria, multidisciplinary field team, initial home visit to address unmet SDOH.	To reduce 30-day unplanned hospital readmissions, emergency department, and 9-1-1 system use, and overall health care cost.	230 patients have been successfully discharged from the program. Patient satisfaction scores are > 90%, and > 90% of the patient care is delivered by NPs. Preliminary results suggest that ED use decreased both 1 month and 3 months after enrollment, whereas outpatient use increased.	Formal evaluation of program not discussed as still underway at time of publication.	This CP-THS program shows the potential to reduce costs and improve the quality of care patients receive. Addressing SDOH requires a multidisciplinary approach.	Low- Implementation of new program not evaluated.
12	Gonzalo et al. 2019	NA Opinion Piece	NA	NA	NA	NA	NA	NA	NA	Authors examine new competencies that medical education must incorporate to better align the education and clinical mission of AHCs, and ensure that the next generation of physician can lead system transformation.	NA - Opinion piece

13	Graham et al 2015	Retrospective cohort study	Urban academic medical center - all admissions from the hospital's primary care network from 1 January 2009 to 31 December 2010.	N = 13,334 admissions, representing 8078 patients)	Data sources included the hospital's clinical and administrative data repositories; professional fee billing repository; electronic provider order entry data; and electronic initial nursing assessment tool reviewed to collect clinical and socioeconomic data. Variables were grouped into two categories as related to early or late readmission.	NA	Early readmissions (readmissions within 7 days after discharge from the index hospitalization), and late readmissions (readmissions 8 to 30 days after discharge)	Overall 30-day readmission rate was 19.7%. 1046 were early readmissions (7.8% of index admissions) and 1586 were late readmissions (11.9% of index admissions). Early readmissions were associated with markers of acute illness burden. Early readmissions were less likely if a patient was discharged in the morning. Late readmissions were associated with markers of chronic illness burden, including receiving a medication indicating organ failure or hemodialysis and social determinants of health, including barriers to learning and having supplemented Medicare or Medicaid.	Only readmissions at one institution were included.	Readmissions within 7 days after discharge were associated with variables related to illness acuity and inpatient care processes, whereas those occurring later were associated with markers of chronic disease burden. Therefore different efforts may be needed to prevent early vs. late readmissions.	Moderate - Large sample size retrospective cohort study. Findings not generalizable.
14	Hatzi et al 2020	Descriptive analysis	Johns Hopkins Health System (JHHS).	N= 1,094,683	Data about housing issues extracted from the electronic health record (EHR), both structured and unstructured, and compared this to community-level characteristics of patients' neighborhood as measured by the Area Deprivation Index.	Housing issues identified as a proxy for individual-level social needs.	Evaluation of individual- and community-level needs of housing (one of the SOOH domains) and exploration how the level of housing needs impacted health care utilization.	Depended on EHR data and use of ICD codes. Focused on data before 2018. Utilization rates measured by encounters in EHR at JHHS and may have missed other types of encounters or encounters outside the system.	Study findings can serve as baseline data and can inform health care systems aiming to use their EHRs to capture social needs and integrate them into patient care and population health management interventions.	Moderate - Large sample size descriptive analysis dependent on EHR data and ICD codes.	
15	Horwitz et al. 2020	Literature search.	US health systems identified in literature search	NA - literature search	LexisNexis and Google search for news articles and press releases that included the terms health system or hospital and investment and social determinant of health or community health in the two-year period January 1, 2017 – November 30, 2019, with the US as the location.	Search of public announcements.	Health Systems financial investments in SOOH	Health systems may have made investments without publicly announcing them and search strategy may have missed some announcements, investments may have been overestimated and data was unable to be at the level of individual hospitals within a health system.	Health systems are making large investments in SOOH, especially on housing but this is still very small fraction of overall spending.	Low- search of public announcements.	
16	Kaplan et al. 2020	NA Case Study	New York University Langone Health (NYU-LH)	NA	NA	NYU-LH partnered with local CBOs to design, implement, and evaluate several community health improvement projects.	NA	Authors describe two community improvement projects - Parent Corps and Tobacco Free Community.	Challenges included agreeing on selected projects and building trusting relations with CBOs.	Authors portray this case study as a new paradigm for health system investment in area-wide population health improvement through partnering with CBOs to oversee the Community Health Needs Assessment (CHNA).	Low- Case Study describing two improvement projects

17	Kapf 2019	NA Opinion piece	NA Opinion piece	NA	NA	NA	NA	NA	NA	NA	NA	NA	Argues that providing affordable and comprehensive health coverage for all is a necessary first step and federal, state and local governments have the necessary resources to address poverty and the roots of the SDOH.	NA - Opinion piece
18	Knettel 2011	NA Opinion piece	NA Opinion piece	NA	NA	NA	NA	NA	NA	NA	NA	NA	Business case for AHC's to address environmental, social, and behavioral determinants of health because the fear that addressing SDOH will result in declining revenues is overstated.	NA - Opinion piece
19	Leider et al. 2017	Financial analysis.	Not-for-profit hospitals	2608 hospitals (2009), 2560 hospitals (2010), 2016 hospitals (2011), 1215 hospitals (2012)	Using data from 2009 to 2012 hospital tax and other governmental filings; national, hospital referral-region, and facility-level estimates of Community Benefit spending were constructed. Data were matched at the facility level for a non-profit hospital's IRS tax filings (Form 990, Schedule H) and CMS Hospital Cost Report Information System and Provider of Service data sets.	NA	NA	NA	Descriptive analysis of financial data	Community Benefit spending	In 2009, hospital systems spent slightly less than 8% of their total operating expenses on Community Benefits, on average. This increased to 8.3% to 8.5% in 2012. > 80% of this went toward charity care, unreimbursed Medicaid, and subsidized health services, with approximately 6% going toward both community health improvement and health professionals' education.	Lack of availability of data for 2011 and 2012, due to IRS filing rules. Inability to extrapolate data to population based activities.	Low- descriptive financial analysis study.	
20	Lin et al 2017	RCT	Patients in an urban AHC (Brigham and Women's Hospital)	N=72	After 7 months, ED visits, hospitalizations, and costs for the intervention and control groups were analyzed.	NA	NA	NA	A community health worker (CHW) who assisted patients with unmet social needs, and a team that developed an interdisciplinary care plan for eligible patients.	ED visits, admissions and costs.	Intervention group had 35% lower ED visits, 31% fewer admissions, and 15% lower costs than the control group	Results are not statistically significant due to the small sample size	Moderate - RCT with small sample size and not statistically significant results.	
21	Meyer et al. 2020	Implementation, and feasibility study of universal screening for SDOH using PDSA cycles.	Four hospital-affiliated, community based primary care practices at New York Presbyterian Hospital at Columbia University.	N= 13273	Multidisciplinary teams performing SDOH and clinical screening and customized referrals for services.	NA	NA	NA	Screenings completed on tablets via NowPow (an SDOH screening and referral platform), in addition to clinical screening (e.g. drug use, alcohol use, asthma).	Screening for food insecurity, housing insecurity, transportation needs, utility needs, and safety needs.	1939 patients identified with previously undetected needs, and 944 were enrolled in navigation to address social service needs. Screening rates were 27% (food insecurity), 25% (housing insecurity), 12% (transportation needs), 8% (utility needs), and 1% (safety needs).	Added technological burdens.	Low- Implementation study using PDSA cycles.	
22	Michener et al. 2012	NA Opinion Piece	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	Authors encourage academic health centers (AHCs) to better engage with their communities and build a community-engaged research	NA - Opinion piece

23	Navathe et al 2018	Observational study	Patients with cardiovascular disease admitted from January 1, 2011, to December 31, 2013 at an academic health system in southeastern Massachusetts	N=49319	Evaluation of payer claims, EHR data, and physician notes extracted from a centralized clinical registry, using multivariable logistic regression to adjust for patient characteristics.	Evaluate the prevalence of seven social factors using physician notes as compared to claims and structured electronic health records (EHRs) data and the resulting association with 30-day readmissions.	All-payer claims, EHR data, and physician notes extracted from a centralized clinical registry for the seven social factors: tobacco use, alcohol abuse, drug abuse, depression, housing instability, fall risk, and poor social support.	All seven social characteristics were identified at the highest rates in physician notes. Compared to an 18.6 percent baseline readmission rate, risk-adjusted analysis showed higher readmission risk for patients with housing instability (readmission rate 24.5 percent; $p < .001$), depression (20.6 percent; $p < .001$), drug abuse (20.2 percent; $p = .01$), and poor social support (20.0 percent; $p = .01$).	Study is limited to an estimate of the association between social factors and readmission and not the causal relationship does not account for admissions to other hospitals, results may not be generalizable to other institutions.	The seven social risk factors studied are substantially more prevalent than represented in administrative data. Automated methods for analyzing physician notes may enable better identification of patients with social needs.	Moderate - Observational study limited to an estimate of the association between social factors and readmission. Large sample size
24	Ostra 2018	NA Case study	The University of Toledo College of Medicine partnering with ProMedica in Ohio	NA	NA	Proposed large-scale economic development to address the SDOH through a 50-year, \$2.5 billion relationship	NA	Neighborhood redevelopment plans, the development of the ProMedica National Center for Social Determinants of Health Research and the recruitment of new faculty to lead this transformation.	NA	Authors believe current steps and long term strategy will have long lasting effects.	Low - single case study
25	Park et al 2019	NA Opinion piece	NA	NA	NA	NA	NA	NA	NA	Argues that AHC's focus their missions on social determinants to create partnerships with their communities called "academic-community health systems"	NA - Opinion piece
26	Perman et al 2015	NA	NA Opinion piece	NA	NA	NA	NA	NA	NA	Argue for "infusing" SDOH in all aspects of AHC clinical care, research and education through community partnerships	NA - Opinion piece
27	Rieselbach et al. 2018	NA	NA Opinion piece	NA	NA	NA	NA	NA	NA	Authors argue that a new paradigm where Community Health Centers (CHCs) partner with AHCs represents a potential strategy to serve more Medicaid patients.	NA - Opinion piece
28	Rosenbaum et al. 2015	Financial analysis.	NA - analysis of financial data.	2,980 private nonprofit hospitals.	Analysis of four tax expenditure categories for each hospital: federal income tax, state income tax, state and local sales taxes, and local property tax. Also combined values for charitable contributions and tax exempt bonds and aggregated totals at the state level using Joint Committee on Taxation's methodology.	Analysis of 2011 cost reports from the Healthcare Cost Report Information System (HCRIS) maintained by the Centers for Medicare and Medicaid Services and the 2011 Form 990s maintained by the IRS.	Total value of the nonprofit tax exemption.	Total tax exemption reached \$24.6 billion in 2011. Corporate income tax (federal) \$6.3 billion Corporate income tax (state) \$1.2 billion, Tax-exempt bond financing (federal) \$3.3 billion Charitable contributions (federal) \$3.4 billion Sales tax (state and local) \$6.1 billion Property tax (local) \$4.3 billion.	Potential errors in the reported financial data, some assumption used may slightly alter the data and the potential for some facilities mistakenly added.	There is a substantial public investment in tax-exempt hospitals, and therefore a public interest not in community benefits and how they are allocated. The authors hope that hospitals expand their mission to include community health improvement and subsequently improve overall health, while reducing subpopulation health disparities.	Low-descriptive financial analysis study.
29	Rubin et al 2014	Qualitative study	Patients readmitted to the hospital within 30 days of discharge, who had diabetes in an urban academic medical center in Philadelphia.	N=20	Thematic analysis of semi-structured interviews.	Three investigators performed the interviews which were recorded and transcribed.	To explore causes of early readmission, i.e., hospital readmission within 30 days of discharge, among patients with diabetes.	Five themes emerged: (1) poor health literacy, (2) health system failure (of the discharge process and post-discharge support), (3) failure of expected protective factors, (e.g., following the discharge instructions), (4) social determinants of health impeding care, and (5) loss of control over illness.	Single hospital using individual control group and multiple patients had readmissions which may have affected recall.	Interventions such as education and clear discharge communication may prevent readmission in patients with diabetes.	Low - single hospital with no control group. Not generalizable
30	Schroeder 2016	NA Opinion piece	NA Opinion piece	NA	NA	NA	NA	NA	NA	The essential role of social justice as the "moral core" and expressing solidarity with the community	NA - Opinion piece

31	Smithman et al 2019	NA Opinion piece	NA Opinion piece	NA	NA	NA	NA	NA	NA	NA	NA	Appeals to make social accountability a fourth mission for AHCs, in addition to the traditional missions of education, research and clinical care	NA	Moderate- Real world controlled clinical trial with controls possibly exposed to the intervention.
32	Tung et al 2018	Real-world controlled clinical trial with participants assigned by alternating week to receive the intervention or usual care	Publicly insured adults, ages 45– 74 years at an urban academic medical center (University of Chicago) in 2015–2016	N = 374	Surveys were administered at baseline, 1 week, 1 month, and 3 months.	CommunityRx generated an automated, personalized list of resources, known as HealthRx, near each participant's home using condition-specific, evidence-based algorithms.	impact of HealthRx referrals on (1) knowledge of the most commonly referred resource types (e.g. healthy eating classes), and (2) beliefs about having resources in the community to manage health.	intervention recipients improved knowledge (AOR = 2.15; 95% CI, 1.29–3.58) and beliefs (AOR= 1.68; 95% CI, 1.07–2.64) about common resources in the community to manage health: smoking cessation (AOR = 2.76; 95% CI, 1.07–7.12) and weight loss resources (AOR = 2.26; 95% CI 1.05–4.84). Positive changes associated with higher resource use (P= 0.02).	Some controls possibly exposed to the intervention in this real world trial. Participants may have over- or underestimated their knowledge of local resources. Unable to link the specific HealthRx generated for each patient to their survey responses.	HealthRx improved both knowledge and overall beliefs about community resources in a highly vulnerable population. Using low-intensity health IT intervention to deliver resource referrals may lead to better knowledge retention and beliefs about resources in the local community.	Appeals to make social accountability a fourth mission for AHCs, in addition to the traditional missions of education, research and clinical care	NA	Moderate- Real world controlled clinical trial with controls possibly exposed to the intervention.	
33	Wartman et al 2011	NA Opinion piece	NA	NA	NA	NA	NA	NA	NA	NA	Ague for a nation-wide movement of AHCs to address SDOH collectively. Authors propose a new model where multi-disciplinary teams in the areas of education, research and clinical care across departments have a specific area of focus and the creation of a compact between AHCs and community leaders	NA - Opinion piece		
34	Wilkins et al 2019	NA Opinion piece	NA	NA	NA	NA	NA	NA	NA	NA	Community engagement and integration is viewed as a vital component to help AHCs better understand and address SDOH	NA - Opinion piece		
35	Wu et al 2019	Cluster RCT	22 CBOs located in Baltimore randomly assigned other control or intervention group.	N=5,255 patients	JHHS partnered with 22 CBOs randomly assigned to the intervention or control group	An online tool to refer clients to community resources and meet and greet sessions for CBO staff with healthcare staff	Primary outcomes were emergency department visits and days in the hospital. Additional outcomes for CBO clients (included knowledge of other CBOs), and outcomes for CBO staff included the number of referrals made to and received from the healthcare system and the number of referrals made to and received from other CBOs.	2.9% increase in referrals by inpatient staff to intervention CBOs (p =0.051). 6.6% increase in referrals by outpatient staff to intervention CBOs between baseline and follow-up (p =0.027). Outpatient staff reported a significant reduction in barriers related to the lack of information about CBO services (118.3% p =0.004) and an increase in confidence in community resources (+14.4%, p =0.023) from baseline to follow-up.	Inability to link patients to specific CBOs due to improved privacy concerns, outcomes depended on changes in organization strategies, intervention period may have been too short to build trust with CBOs.	The intervention did not improve healthcare utilization outcomes but did increase staff knowledge and confidence in local CBOs.	Moderate- Cluster RCT design with large sample size			

36	Wynn et al. 2020	Quality initiative pilot with a pre- and post design.	3 acute care units at a large urban academic medical center.	2354 patients admitted over a 3 month period	All patients admitted to these units were eligible to be screened for food insecurity with education was offered to nurses who provided patient care on the pilot units.	Patients screened for SDOH needs, if identified as food insecure, linked to community resources and provided with a bag of food on discharge. Education offered to nurses and a pre- and post survey was administered to assess SDOH knowledge and confidence.	Patients experiencing food insecurity via ENHR report and knowledge of nurses about ability and knowledge to discuss food access with patients.	2063 (88%) were screened for SDOH and 220 (10%) were positive for food insecurity. 97% of nurses participated in education and demonstrated increased knowledge and confidence ($P < .001$).	Post survey lower than expected and additional burden to nursing workload.	Food insecurity is prevalent amongst patients and this Food as medicine program is feasible.	Low - Pilot QI study with moderate sample size
37	Young et al. 2013	Financial cross-sectional analysis.	Private tax exempt hospitals	1800 hospitals	Form 990 and the related Schedule H for 2009 through Guidestar. Descriptive statistics for each of the community-benefit measures were used. For the analytic models, we used two multiple-regression models, one for each type of community-benefit measure: patient care and community service. For the regression models, the independent variables comprised institutional-level, community-level, and market-level characteristics.	Combined financial data with other data and used descriptive statistics.	Seven community benefits: Charity care, Unreimbursed costs for means-tested government programs, Subsidized health services, Community health improvement, Cash or in-kind contributions to community groups, Research and Health professions education.	Hospitals expended, on average, 7.5% of their operating expenses for these services and activities but there was a large range with hospitals in the top decile spent an average of 20.1%, whereas those in the bottom decile spent an average of 1.1%. More than 85% of these expenses were devoted to charity care and other patient care services. Of the remaining community-benefit expenditures, approximately 5% were devoted to community health improvements that hospitals undertook directly.	Unable to determine the causal connection between the reporting requirements and provision of benefits.	Authors concluded that tax-exempt hospitals varied markedly in the level of community benefits provided. The most of their benefit-related expenditures were related to patient care services and little was spent on community health improvement.	Low-descriptive financial analysis study.
38	Zhang et al 2020	Retrospective cohort study	Electronic health record data for hospital admissions between January 2015 and November 2017 at an academic medical center in New York City.	N = 19,941 hospital admissions	Simplified HOSPITAL score model to predict potentially avoidable 30-day readmission or death. Analysis was conducted for all patients and three subgroups: 1) Medicaid patients, 2) patients who are 65 or older, and 3) obese patients	NA - extracted data from EHR	To examine the impact of social determinants of health on predictive models for potentially avoidable 30-day readmission.	Adding SDOH did not improve the prediction among all patients. Adding individual- and community-level SDOH at the US census tract level significantly improved the prediction for all three subgroups. Specifically, C-statistics improved from 0.70 to 0.73 for Medicaid patients, from 0.66 to 0.68 for patients 65 or older, and from 0.70 to 0.73 for obese patients.	EHR data from a single academic medical center. Results may not be generalizable. Some individual level SDOH such as income were not available. Post discharge mortality with EHR.	Patients from the subgroups studied may be more likely to be affected by social risk factors. Incorporating SDOH into predictive models may be helpful to identify these patients and reduce health disparities.	Moderate - Retrospective cohort study with large sample size. Conducted at single academic medical center with low generalizability.

APPENDIX B: HOSPITAL FORMS

SCHEDULE H (Form 990)

Department of the Treasury
Internal Revenue Service

Hospitals

- ▶ **Complete if the organization answered "Yes" on Form 990, Part IV, question 20.**
▶ **Attach to Form 990.**
▶ **Go to www.irs.gov/Form990 for instructions and the latest information.**

OMB No. 1545-0047

2019

**Open to Public
Inspection**

Name of the organization

Employer identification number

Part I Financial Assistance and Certain Other Community Benefits at Cost

		Yes	No
1a Did the organization have a financial assistance policy during the tax year? If "No," skip to question 6a	1a		
b If "Yes," was it a written policy?	1b		
2 If the organization had multiple hospital facilities, indicate which of the following best describes application of the financial assistance policy to its various hospital facilities during the tax year. ☐ Applied uniformly to all hospital facilities ☐ Applied uniformly to most hospital facilities ☐ Generally tailored to individual hospital facilities			
3 Answer the following based on the financial assistance eligibility criteria that applied to the largest number of the organization's patients during the tax year.			
a Did the organization use Federal Poverty Guidelines (FPG) as a factor in determining eligibility for providing <i>free</i> care? If "Yes," indicate which of the following was the FPG family income limit for eligibility for free care: ☐ 100% ☐ 150% ☐ 200% ☐ Other _____ %	3a		
b Did the organization use FPG as a factor in determining eligibility for providing <i>discounted</i> care? If "Yes," indicate which of the following was the family income limit for eligibility for discounted care: ☐ 200% ☐ 250% ☐ 300% ☐ 350% ☐ 400% ☐ Other _____ %	3b		
c If the organization used factors other than FPG in determining eligibility, describe in Part VI the criteria used for determining eligibility for free or discounted care. Include in the description whether the organization used an asset test or other threshold, regardless of income, as a factor in determining eligibility for free or discounted care.			
4 Did the organization's financial assistance policy that applied to the largest number of its patients during the tax year provide for free or discounted care to the "medically indigent"?	4		
5a Did the organization budget amounts for free or discounted care provided under its financial assistance policy during the tax year?	5a		
b If "Yes," did the organization's financial assistance expenses exceed the budgeted amount?	5b		
c If "Yes" to line 5b, as a result of budget considerations, was the organization unable to provide free or discounted care to a patient who was eligible for free or discounted care?	5c		
6a Did the organization prepare a community benefit report during the tax year?	6a		
b If "Yes," did the organization make it available to the public?	6b		

Complete the following table using the worksheets provided in the Schedule H instructions. Do not submit these worksheets with the Schedule H.

7 Financial Assistance and Certain Other Community Benefits at Cost

	(a) Number of activities or programs (optional)	(b) Persons served (optional)	(c) Total community benefit expense	(d) Direct offsetting revenue	(e) Net community benefit expense	(f) Percent of total expense
Financial Assistance and Means-Tested Government Programs						
a Financial Assistance at cost (from Worksheet 1)						
b Medicaid (from Worksheet 3, column a)						
c Costs of other means-tested government programs (from Worksheet 3, column b)						
d Total. Financial Assistance and Means-Tested Government Programs						
Other Benefits						
e Community health improvement services and community benefit operations (from Worksheet 4)						
f Health professions education (from Worksheet 5)						
g Subsidized health services (from Worksheet 6)						
h Research (from Worksheet 7)						
i Cash and in-kind contributions for community benefit (from Worksheet 8)						
j Total. Other Benefits						
k Total. Add lines 7d and 7j						

For Paperwork Reduction Act Notice, see the Instructions for Form 990.

Cat. No. 50192T

Schedule H (Form 990) 2019

Part II Community Building Activities Complete this table if the organization conducted any community building activities during the tax year, and describe in Part VI how its community building activities promoted the health of the communities it serves.

	(a) Number of activities or programs (optional)	(b) Persons served (optional)	(c) Total community building expense	(d) Direct offsetting revenue	(e) Net community building expense	(f) Percent of total expense
1 Physical improvements and housing						
2 Economic development						
3 Community support						
4 Environmental improvements						
5 Leadership development and training for community members						
6 Coalition building						
7 Community health improvement advocacy						
8 Workforce development						
9 Other						
10 Total						

Part III Bad Debt, Medicare, & Collection Practices

Section A. Bad Debt Expense

- 1** Did the organization report bad debt expense in accordance with Healthcare Financial Management Association Statement No. 15? **1**
- 2** Enter the amount of the organization's bad debt expense. Explain in Part VI the methodology used by the organization to estimate this amount **2**
- 3** Enter the estimated amount of the organization's bad debt expense attributable to patients eligible under the organization's financial assistance policy. Explain in Part VI the methodology used by the organization to estimate this amount and the rationale, if any, for including this portion of bad debt as community benefit. **3**
- 4** Provide in Part VI the text of the footnote to the organization's financial statements that describes bad debt expense or the page number on which this footnote is contained in the attached financial statements.

Section B. Medicare

- 5** Enter total revenue received from Medicare (including DSH and IME) **5**
- 6** Enter Medicare allowable costs of care relating to payments on line 5 **6**
- 7** Subtract line 6 from line 5. This is the surplus (or shortfall) **7**
- 8** Describe in Part VI the extent to which any shortfall reported on line 7 should be treated as community benefit. Also describe in Part VI the costing methodology or source used to determine the amount reported on line 6. Check the box that describes the method used:
- ☐ Cost accounting system ☐ Cost to charge ratio ☐ Other

Section C. Collection Practices

- 9a** Did the organization have a written debt collection policy during the tax year? **9a**
- b** If "Yes," did the organization's collection policy that applied to the largest number of its patients during the tax year contain provisions on the collection practices to be followed for patients who are known to qualify for financial assistance? Describe in Part VI **9b**

Part IV Management Companies and Joint Ventures (owned 10% or more by officers, directors, trustees, key employees, and physicians—see instructions)

(a) Name of entity	(b) Description of primary activity of entity	(c) Organization's profit % or stock ownership %	(d) Officers, directors, trustees, or key employees' profit % or stock ownership %	(e) Physicians' profit % or stock ownership %
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				

Part V Facility Information**Section A. Hospital Facilities**

(list in order of size, from largest to smallest—see instructions)
How many hospital facilities did the organization operate during the tax year? _____

Name, address, primary website address, and state license number (and if a group return, the name and EIN of the subordinate hospital organization that operates the hospital facility)

	Licensed hospital	General medical & surgical	Children's hospital	Teaching hospital	Critical access hospital	Research facility	ER-24 hours	ER—other	Other (describe)	Facility reporting group
1										
2										
3										
4										
5										
6										
7										
8										
9										
10										

Part V Facility Information (continued)**Section B. Facility Policies and Practices**

(complete a separate Section B for each of the hospital facilities or facility reporting groups listed in Part V, Section A)

Name of hospital facility or letter of facility reporting group _____

Line number of hospital facility, or line numbers of hospital facilities in a facility reporting group (from Part V, Section A): _____

	Yes	No
Community Health Needs Assessment		
1 Was the hospital facility first licensed, registered, or similarly recognized by a state as a hospital facility in the current tax year or the immediately preceding tax year?	1	
2 Was the hospital facility acquired or placed into service as a tax-exempt hospital in the current tax year or the immediately preceding tax year? If "Yes," provide details of the acquisition in Section C	2	
3 During the tax year or either of the two immediately preceding tax years, did the hospital facility conduct a community health needs assessment (CHNA)? If "No," skip to line 12	3	
If "Yes," indicate what the CHNA report describes (check all that apply):		
a ☐ A definition of the community served by the hospital facility		
b ☐ Demographics of the community		
c ☐ Existing health care facilities and resources within the community that are available to respond to the health needs of the community		
d ☐ How data was obtained		
e ☐ The significant health needs of the community		
f ☐ Primary and chronic disease needs and other health issues of uninsured persons, low-income persons, and minority groups		
g ☐ The process for identifying and prioritizing community health needs and services to meet the community health needs		
h ☐ The process for consulting with persons representing the community's interests		
i ☐ The impact of any actions taken to address the significant health needs identified in the hospital facility's prior CHNA(s)		
j ☐ Other (describe in Section C)		
4 Indicate the tax year the hospital facility last conducted a CHNA: 20 _____		
5 In conducting its most recent CHNA, did the hospital facility take into account input from persons who represent the broad interests of the community served by the hospital facility, including those with special knowledge of or expertise in public health? If "Yes," describe in Section C how the hospital facility took into account input from persons who represent the community, and identify the persons the hospital facility consulted	5	
6a Was the hospital facility's CHNA conducted with one or more other hospital facilities? If "Yes," list the other hospital facilities in Section C	6a	
6b Was the hospital facility's CHNA conducted with one or more organizations other than hospital facilities? If "Yes," list the other organizations in Section C	6b	
7 Did the hospital facility make its CHNA report widely available to the public?	7	
If "Yes," indicate how the CHNA report was made widely available (check all that apply):		
a ☐ Hospital facility's website (list url): _____		
b ☐ Other website (list url): _____		
c ☐ Made a paper copy available for public inspection without charge at the hospital facility		
d ☐ Other (describe in Section C)		
8 Did the hospital facility adopt an implementation strategy to meet the significant community health needs identified through its most recently conducted CHNA? If "No," skip to line 11	8	
9 Indicate the tax year the hospital facility last adopted an implementation strategy: 20 _____		
10 Is the hospital facility's most recently adopted implementation strategy posted on a website?	10	
a If "Yes," (list url): _____		
b If "No," is the hospital facility's most recently adopted implementation strategy attached to this return?	10b	
11 Describe in Section C how the hospital facility is addressing the significant needs identified in its most recently conducted CHNA and any such needs that are not being addressed together with the reasons why such needs are not being addressed.		
12a Did the organization incur an excise tax under section 4959 for the hospital facility's failure to conduct a CHNA as required by section 501(r)(3)?	12a	
b If "Yes" to line 12a, did the organization file Form 4720 to report the section 4959 excise tax?	12b	
c If "Yes" to line 12b, what is the total amount of section 4959 excise tax the organization reported on Form 4720 for all of its hospital facilities? \$ _____		

Part V Facility Information *(continued)***Financial Assistance Policy (FAP)**

Name of hospital facility or letter of facility reporting group _____

		Yes	No
Did the hospital facility have in place during the tax year a written financial assistance policy that:			
13	Explained eligibility criteria for financial assistance, and whether such assistance included free or discounted care? If "Yes," indicate the eligibility criteria explained in the FAP:	13	
a	☐ Federal poverty guidelines (FPG), with FPG family income limit for eligibility for free care of _____ % and FPG family income limit for eligibility for discounted care of _____ %		
b	☐ Income level other than FPG (describe in Section C)		
c	☐ Asset level		
d	☐ Medical indigency		
e	☐ Insurance status		
f	☐ Underinsurance status		
g	☐ Residency		
h	☐ Other (describe in Section C)		
14	Explained the basis for calculating amounts charged to patients?	14	
15	Explained the method for applying for financial assistance? If "Yes," indicate how the hospital facility's FAP or FAP application form (including accompanying instructions) explained the method for applying for financial assistance (check all that apply):	15	
a	☐ Described the information the hospital facility may require an individual to provide as part of his or her application		
b	☐ Described the supporting documentation the hospital facility may require an individual to submit as part of his or her application		
c	☐ Provided the contact information of hospital facility staff who can provide an individual with information about the FAP and FAP application process		
d	☐ Provided the contact information of nonprofit organizations or government agencies that may be sources of assistance with FAP applications		
e	☐ Other (describe in Section C)		
16	Was widely publicized within the community served by the hospital facility? If "Yes," indicate how the hospital facility publicized the policy (check all that apply):	16	
a	☐ The FAP was widely available on a website (list url): _____		
b	☐ The FAP application form was widely available on a website (list url): _____		
c	☐ A plain language summary of the FAP was widely available on a website (list url): _____		
d	☐ The FAP was available upon request and without charge (in public locations in the hospital facility and by mail)		
e	☐ The FAP application form was available upon request and without charge (in public locations in the hospital facility and by mail)		
f	☐ A plain language summary of the FAP was available upon request and without charge (in public locations in the hospital facility and by mail)		
g	☐ Individuals were notified about the FAP by being offered a paper copy of the plain language summary of the FAP, by receiving a conspicuous written notice about the FAP on their billing statements, and via conspicuous public displays or other measures reasonably calculated to attract patients' attention		
h	☐ Notified members of the community who are most likely to require financial assistance about availability of the FAP		
i	☐ The FAP, FAP application form, and plain language summary of the FAP were translated into the primary language(s) spoken by Limited English Proficiency (LEP) populations		
j	☐ Other (describe in Section C)		

Schedule H (Form 990) 2019

Part V Facility Information (continued)**Billing and Collections**

Name of hospital facility or letter of facility reporting group _____

	Yes	No
17 Did the hospital facility have in place during the tax year a separate billing and collections policy, or a written financial assistance policy (FAP) that explained all of the actions the hospital facility or other authorized party may take upon nonpayment?	17	
18 Check all of the following actions against an individual that were permitted under the hospital facility's policies during the tax year before making reasonable efforts to determine the individual's eligibility under the facility's FAP:		
a ☐ Reporting to credit agency(ies)		
b ☐ Selling an individual's debt to another party		
c ☐ Deferring, denying, or requiring a payment before providing medically necessary care due to nonpayment of a previous bill for care covered under the hospital facility's FAP		
d ☐ Actions that require a legal or judicial process		
e ☐ Other similar actions (describe in Section C)		
f ☐ None of these actions or other similar actions were permitted		
19 Did the hospital facility or other authorized party perform any of the following actions during the tax year before making reasonable efforts to determine the individual's eligibility under the facility's FAP?	19	
If "Yes," check all actions in which the hospital facility or a third party engaged:		
a ☐ Reporting to credit agency(ies)		
b ☐ Selling an individual's debt to another party		
c ☐ Deferring, denying, or requiring a payment before providing medically necessary care due to nonpayment of a previous bill for care covered under the hospital facility's FAP		
d ☐ Actions that require a legal or judicial process		
e ☐ Other similar actions (describe in Section C)		
20 Indicate which efforts the hospital facility or other authorized party made before initiating any of the actions listed (whether or not checked) in line 19 (check all that apply):		
a ☐ Provided a written notice about upcoming ECAs (Extraordinary Collection Action) and a plain language summary of the FAP at least 30 days before initiating those ECAs (if not, describe in Section C)		
b ☐ Made a reasonable effort to orally notify individuals about the FAP and FAP application process (if not, describe in Section C)		
c ☐ Processed incomplete and complete FAP applications (if not, describe in Section C)		
d ☐ Made presumptive eligibility determinations (if not, describe in Section C)		
e ☐ Other (describe in Section C)		
f ☐ None of these efforts were made		

Policy Relating to Emergency Medical Care

21 Did the hospital facility have in place during the tax year a written policy relating to emergency medical care that required the hospital facility to provide, without discrimination, care for emergency medical conditions to individuals regardless of their eligibility under the hospital facility's financial assistance policy?	21	
If "No," indicate why:		
a ☐ The hospital facility did not provide care for any emergency medical conditions		
b ☐ The hospital facility's policy was not in writing		
c ☐ The hospital facility limited who was eligible to receive care for emergency medical conditions (describe in Section C)		
d ☐ Other (describe in Section C)		

Part V Facility Information *(continued)***Charges to Individuals Eligible for Assistance Under the FAP (FAP-Eligible Individuals)**

Name of hospital facility or letter of facility reporting group _____

	Yes	No
22 Indicate how the hospital facility determined, during the tax year, the maximum amounts that can be charged to FAP-eligible individuals for emergency or other medically necessary care.		
a ☐ The hospital facility used a look-back method based on claims allowed by Medicare fee-for-service during a prior 12-month period		
b ☐ The hospital facility used a look-back method based on claims allowed by Medicare fee-for-service and all private health insurers that pay claims to the hospital facility during a prior 12-month period		
c ☐ The hospital facility used a look-back method based on claims allowed by Medicaid, either alone or in combination with Medicare fee-for-service and all private health insurers that pay claims to the hospital facility during a prior 12-month period		
d ☐ The hospital facility used a prospective Medicare or Medicaid method		
23 During the tax year, did the hospital facility charge any FAP-eligible individual to whom the hospital facility provided emergency or other medically necessary services more than the amounts generally billed to individuals who had insurance covering such care?	23	
If "Yes," explain in Section C.		
24 During the tax year, did the hospital facility charge any FAP-eligible individual an amount equal to the gross charge for any service provided to that individual?	24	
If "Yes," explain in Section C.		

Schedule H (Form 990) 2019

Part V **Facility Information** *(continued)*

Section C. Supplemental Information for Part V, Section B. Provide descriptions required for Part V, Section B, lines 2, 3j, 5, 6a, 6b, 7d, 11, 13b, 13h, 15e, 16j, 18e, 19e, 20a, 20b, 20c, 20d, 20e, 21c, 21d, 23, and 24. If applicable, provide separate descriptions for each hospital facility in a facility reporting group, designated by facility reporting group letter and hospital facility line number from Part V, Section A ("A, 1," "A, 4," "B, 2," "B, 3," etc.) and name of hospital facility.

This image shows a full page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the page.

Part V Facility Information *(continued)***Section D. Other Health Care Facilities That Are Not Licensed, Registered, or Similarly Recognized as a Hospital Facility**
(list in order of size, from largest to smallest)

How many non-hospital health care facilities did the organization operate during the tax year? _____

Name and address	Type of Facility (describe)
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Part VI Supplemental Information

Provide the following information.

- 1 Required descriptions.** Provide the descriptions required for Part I, lines 3c, 6a, and 7; Part II and Part III, lines 2, 3, 4, 8 and 9b.
- 2 Needs assessment.** Describe how the organization assesses the health care needs of the communities it serves, in addition to any CHNAs reported in Part V, Section B.
- 3 Patient education of eligibility for assistance.** Describe how the organization informs and educates patients and persons who may be billed for patient care about their eligibility for assistance under federal, state, or local government programs or under the organization's financial assistance policy.
- 4 Community information.** Describe the community the organization serves, taking into account the geographic area and demographic constituents it serves.
- 5 Promotion of community health.** Provide any other information important to describing how the organization's hospital facilities or other health care facilities further its exempt purpose by promoting the health of the community (e.g., open medical staff, community board, use of surplus funds, etc.).
- 6 Affiliated health care system.** If the organization is part of an affiliated health care system, describe the respective roles of the organization and its affiliates in promoting the health of the communities served.
- 7 State filing of community benefit report.** If applicable, identify all states with which the organization, or a related organization, files a community benefit report.

[illegible]

APPENDIX C: KEY INFORMANT INTERVIEW GUIDE

Date_____

Worksite_____

Key Informant_____

Introduction

My name is Marc Caruana, and I am an employee of Wake Forest Baptist Health and a doctoral candidate at the University of North Carolina, Chapel Hill. For my dissertation, I am interested in learning how health care organizations are addressing the social determinants of health. The purpose of this interview as part of my research study is to learn how your organization is currently addressing social determinants of health and explore the opportunities and challenges you are currently facing.

This interview is voluntary and not participating will not affect your relationship to your organization. You may withdraw from participation at any time even after you consent to participate. The interview should take about 45 minutes. It will be completely confidential and the information you provide will be released as general themes in a combined summary to protect your identity as a research subject. Digital audio files and transcripts will be confidentially destroyed at the end of the research study.

With your permission I would like to record your interview. May I record your interview?

Do you have any questions about the research study or the interview process? If you have questions or concerns about your rights as a research subject, you may contact the UNC_CH Institutional Review Board at 919-966-3113 or IRB_subjects@unc.edu.

I would like to explain some of the terms I just used in my introduction. For this interview, social determinants of health (SDOH) are reflecting five key areas outlined by Healthy People 2020: Economic Stability, Education, Social and Community Context, Health and Health Care and Neighborhood and Built Environment.”

“Community health is a multi-sector and multi-disciplinary collaborative enterprise that uses public health science, evidence-based strategies, and other approaches to engage and work with communities, in a culturally appropriate manner, to optimize the health and quality of life of all persons who live, work, or are otherwise active in a defined community or communities.” For of this interview, community health will refer to the areas where you currently offer clinic or hospital services.

During the interview I will refer to “immediate community” and “other communities” that are part of your health care system. When I refer to your immediate community, I am referring to the county where the main medical center is located at. Other communities that are part of your health care system refers to other counties where other hospitals or clinics within your system are located at.

Introduction questions:

What is your job title? Can you tell me a little bit about your role?

How many years have you been with this organization?

What were some of your other roles at previous organizations?

Community benefit report/financial data question:

How does your organization make decisions on areas for investment; specifically, where to invest and where not to invest?

Based on the most recent community benefit reports, your financial contributions at the primary medical center have been primarily towards _____. Why has your organization primarily contributed to _____?

Can you tell me more about the other areas where you also made contributions?

Social determinants of health questions:

I will ask about each SDOH defined by the Healthy People 2020 framework. As you reflect on these questions, please focus on community-based strategies and community building activities that go beyond patient level strategies based on social risk. I will be asking about both your immediate community and other communities that are part of your health care system as previously defined.

Economic Stability:

How is your organization involved in efforts or initiatives to address economic stability including **poverty, employment, food insecurity** and **housing instability** in your immediate community? What about other communities that are part of your health care system?

Do you see any other opportunities by which your organization can contribute more to increase economic stability in your immediate community? What about other communities that are part of your health care system?

Can you identify current enablers that allow you to integrate economic stability issues as part of how you deliver health care to your immediate community? What about other communities that are part of your health care system?

What are the main challenges you have met or foresee coming across when addressing economic stability issues in your immediate community? What about other communities that are part of your health care system?

Education:

How is your organization involved in education efforts or initiatives to address areas such as **high school graduation rates, enrollment in higher education, language and literacy** and **early childhood education and development** in your immediate community? What about other communities that are part of your health care system?

Do you see any other opportunities for your organization to contribute more to education efforts in your immediate community? Other communities that are part of your health care system?

Can you identify current enablers that allow you to integrate education efforts and initiatives as part of how you deliver health care to your immediate community? What about other communities that are part of your health care system?

What are the main challenges you have met or foresee coming across when addressing education in your immediate community? What about other communities that are part of your health care system?

Health and Health Care:

Is your organization involved in any efforts or initiatives to address **access to primary care** in your immediate community? What about other communities that are part of your health care system?

Do you see any other opportunities for your organization to contribute more to access to health care in your immediate community? What about other communities that are part of your health care system?

Can you identify current enablers that allow you to integrate health care access as part of how you deliver health care to your immediate community? What about other communities that are part of your health care system?

What are the main challenges you have met or foresee coming across when addressing access to health care in your immediate community? What about other communities that are part of your health care system?

Social and Community Context:

Is your organization involved in any efforts or initiatives to address social and community context issues such as social cohesion, civic participation, discrimination, and incarceration in your immediate community? What about other communities that are part of your health care system?
--

Do you see any other opportunities for your organization to contribute more to social and community context issues in your immediate community? What about other communities that are part of your health care system?

Can you identify current enablers that allow you to integrate social and community context issues as part of how you deliver health care to your community? What about other communities that are part of your health care system?

What are the main challenges you have met or foresee when addressing social and community context issues in your immediate community? What about other communities that are part of your health care system?

Neighborhood and Built:

Is your organization involved in any efforts or initiatives to address neighborhood and built environment concerns including access to foods that support healthy eating, quality of housing, crime and violence and environmental conditions in your immediate community? What about other communities that are part of your health care system?
--

Do you see any other opportunities for your organization to contribute to neighborhood and built environment issues in your immediate community? What about other communities that are part of your health care system?
--

Can you identify current enablers that allow you to integrate neighborhood and built environment concerns as part of how you deliver health care to your immediate community? What about other communities that are part of your health care system?

What are the main challenges you have met or foresee coming across when addressing neighborhood and built environment concerns in your immediate community? What about other communities that are part of your health care system?

<u>Closing:</u>

Are there any other additional comments you would like to make about your efforts at addressing SDOH to improve the health of your immediate community or other communities that are part of your health care system?

Is there someone else in your organization that you think will be able to answer some of these questions?

Do you have any other questions for me currently before we conclude?

Probes that may be used throughout:

Can you please expand a little more on this?

Can you please explain what you mean?

Can you please tell me more?

Can you please give me some specific examples.

Provide examples of enablers or challenges e.g. increasing awareness and advocacy, and developing a digital infrastructure.

APPENDIX D: INFORMANT INVITATION EMAIL FOR INTERVIEW

My name is Marc Caruana, and I am the Director of Nurse Anesthesia at Wake Forest Baptist Health and a doctoral candidate at the University of North Carolina, Chapel Hill. This invitation is for a brief interview for a research study that is part of my doctoral dissertation as a DrPH candidate at the UNC Gillings School of Global Public Health.

The purpose of this invitation is to interview you to learn how your organization is currently addressing social determinants of health and explore the opportunities and challenges you are currently facing. I am interested in learning more about your current efforts as an organization regarding addressing the social determinants of health to improve the health of your community.

This interview should take about 45 minutes and can be in person or via a video conference. It will be completely voluntary and confidential and the information you provide will be released as general themes in a combined summary. This worksite will remain blinded and will be listed as Workplace X that is engaged in current Y efforts. With your permission I would like to record your interview. Digital audio files and transcripts will be confidentially destroyed at the end of the research study.

I would like to explain some of the terms I just used in my introduction. When I refer to social determinants of health (SDOH) for the purpose of this study they are defined according to the framework developed Healthy People 2020, which reflects five key areas of social determinants of health: Economic Stability, Education, Social and Community Context, Health and Health Care and Neighborhood and Built Environment.”¹² I am interested in how your organization as a health care provider is interacting with these five key areas in your community.

When I use the term community health, I am referring to it as “Community health is a multi-sector and multi-disciplinary collaborative enterprise that uses public health science, evidence-based strategies, and other approaches to engage and work with communities, in a culturally appropriate manner, to optimize the health and quality of life of all persons who live, work, or are otherwise active in a defined community or communities.” For this interview community health will refer to the areas where you currently offer clinic or hospital services.

Please reply to this email directly if you agree to participate. You are not obligated to participate. If you are unable to participate but know of someone else in your organization who is knowledgeable about this area, please forward me their email address with their permission.

Thank you for your time.

Marc Caruana
DrPH Candidate
UNC Gillings School of Public Health

APPENDIX E: CODEBOOK

Code Name	Description
Challenges	Identified barriers to including SDOH as part of delivering healthcare
Benefits Cliff	Becoming ineligible for public benefits
Community voice	Involving the community in priorities and strategic direction
Digital access and technology	Access to internet and technology including online platforms, telehealth etc.
Lack of trust	Communities or individuals mistrusting authorities
Multi sector	Complex set of factors contributing to challenges
Politics	Intersection of health and politics
Reimbursement and payor mix	Revenue percentages coming from private insurances companies, government insurance programs like Medicare and self-pay patients.
Resources	Supply of capital funding, labor and human resources, information, time etc.
Return on Investment	Demonstration or modeling that investing in new processes, capital etc. will result in improvements in outcomes
Rural areas	Low population density
Segregation and Social Isolation	Separation of different groups by race and ethnicity, and a lack of opportunities to connect with others.
Silos	Lack of communication between teams
Sustainability	The ability to continue to fund projects, new processes etc.
Value based Care	A framework to provide better outcomes for patients at lower overall costs.
Economic Stability	Employment, food insecurity, housing, and poverty

Code Name	Description
Employment	Opportunities for well-paid work
Food Insecurity	Access to affordable healthy food options
Housing	Availability for affordable homes for shelter
Poverty	The inability to fund basic needs including food and shelter
Education	Early childhood and development, enrollment in higher education, high school graduation and language and literacy
Health professions pipeline and primary care	Providing opportunities for a diverse health care workforce and supply of primary care providers.
Language and literacy	Quality and availability of learning and development opportunities for the ability to use words through reading, writing etc.
Workforce development	Ongoing opportunities for training for current and future employees to develop their skills.
Effects of Covid Pandemic	Ramifications and consequences of the Covid pandemic
Focus on disparities	Increased attention to social and economic inequities that result in health disparities between groups and populations
Funding shifts	Changes in funding sources and priorities
Home Health	Providing and supporting health care where a person is living
Increased collaboration	Increased teamwork and strengthening of partnerships.
Telehealth	Health provided without an in-person clinic or health care visit through internet. Phone etc.
Worsening health status	Decline in general health due to worsening medical conditions, functional health etc.

Code Name	Description
Enablers	Identified behaviors, processes, organizational structures etc. that improve the inclusion of SDOH as part of delivering healthcare.
Community partnerships	Working relationships with community-based organizations, religious establishments, local government bodies etc.
Digital Platforms	Online databases that connect individuals and organizations with community resources.
Partnership with local public health	Working relationships with local public health agencies.
Private equity funds and braided funding	Capital investments by private companies and coordinated funding between two or more funding sources.
SDOH Screening	Using screening tools to help identify social and economic needs of individuals that may affect their health outcomes.
Sustained funding	The ability to fund projects and initiatives long term.
Trust and community voice	A relationship built on trust between organizations and communities and the inclusion of ideas for projects and priorities from individuals from the community.
Value based care	A framework to provide better outcomes for patients at lower overall costs
Health and Healthcare	Access to health care, access to primary care and health literacy
Access	The ability to obtain affordable and convenient healthcare services.
APPs and Allied health	Health care providers who are not physicians and other professionals involved with delivering health care services.
Behavioral Health	Health care services for mental health and substance use disorders.
Community Health Workers	A trusted member of a community who improves the quality of care through serving as a link between health care services provided and individuals in the community.

Code Name	Description
Medication access	The ability to obtain necessary medications affordably and conveniently
Telehealth and Home Health	Healthcare provided without an in-person clinic or health care visit through internet. Phone etc. and providing and supporting health care where a person is living.
Wraparound services	Providing non-medical services such as financial assistance, transportation etc. in addition to traditional healthcare services to improve outcomes.
Mission	Evolving tripartite mission of AHCs of education, research, and patient care.
Community Benefit	Activities and programs that provide and improve health benefits to communities to meet their needs.
Healthcare paradigm shift	An evolving approach to the delivery of healthcare that that is moving away from episodic fee for service with a focus on disease towards prevention, factoring in SDOH and value-based care.
Patient Experience	
Neighborhood and Built Environment	Access to foods that support healthy eating patterns, crime and violence, environmental conditions, and quality of housing
Built environment	Buildings including housing, roads, sidewalks etc.
Crime and Gun Violence	Exposure to and prevention efforts to reduce crime and gun violence.
Digital Access	Access to internet and technology such as computers, cell phones etc.
Environmental conditions and sustainability	Exposure to toxic substances, limiting pollution, mitigating the effects of climate change, and making the necessary changes towards a sustainable future.

Code Name	Description
Environmental Justice	Environmental laws and policies that address the disproportionate burden that marginalized communities and the poor experience related climate change, toxic exposure etc.
Food Insecurity	Access to affordable healthy food options
Housing quality	Safe physical housing conditions including clean water, no exposure to toxins in safe environments.
Transportation	A variety of safe affordable, convenient, and affordable options for transport.
Opportunities	Identified areas where AHCs can contribute more
Investing locally	Increasing investments in the local economy.
Trust building	Reach out to certain communities to increase and build trust.
Regulatory changes	New regulations form regulatory bodies like the Joint Commission etc.
Social and Community Context	Civic participation, discrimination, incarceration, and social cohesion
Diversity, Equity, and Inclusion	Organizational structures, programs and activities aimed at improving representation, improving fairness, and imparting a sense of belonging to all the different groups and identities.
Food culture	Beliefs and attitudes and traditions associated with food.
Incarceration	Initiatives for imprisoned individuals and the effects or reentry into society after release.
Social Cohesion	Social support from friends, family, and community-based resources.
Structural racism	Institutional racism embedded in laws, policies and history that perpetuate unfair treatment/ e/g. residential segregation.
Volunteerism and civic participation	Opportunities for teammates to support causes through volunteer opportunities and involvement in local government.

Code Name	Description
Strategy	Overall plan for the organization to achieve long-term goals.
Organizational Structure	A defined hierarchy with identified positions that aligns with the mission and strategy.
Strategic priorities	The objectives outlined by the organization to achieve short and long-term goals.
Teammate Resources	Resources to assist the employees within an organization.

REFERENCES

1. Dzau, V. J. *et al.* Transforming academic health centers for an uncertain future. *N. Engl. J. Med.* **369**, 991–993 (2013).
2. Park, B. *et al.* Health Equity and the Tripartite Mission: Moving From Academic Health Centers to Academic-Community Health Systems. *Acad. Med.* **94**, 1276–1282 (2019).
3. Washington, A. E., Coye, M. J. & Boulware, L. E. Academic health systems' third curve: population health improvement. *JAMA* **315**, 459–460 (2016).
4. *Patient Protection and Affordable Care Act*, 42 U.S.C. § 18001. (2010).
5. The Hilltop Institute University of Maryland, Baltimore County (UMBC). Hospital Community Benefit. *hilltopinstitute.org* at <<https://www.hilltopinstitute.org/our-work/hospital-community-benefit/>>
6. THE AAHC SOCIAL DETERMINANTS OF HEALTH INITIATIVE - Academic Health Centers and the Social Determinants of Health: Challenges & Barriers, Responses & Solutions.
7. Smitherman, H. C., Baker, R. S. & Wilson, M. R. Socially accountable academic health centers: pursuing a quadripartite mission. *Acad. Med.* **94**, 176–181 (2019).
8. Booske, B. C., Athens, J. K., Kindig, D. A., Park, H. & Remington, P. L. COUNTY HEALTH RANKINGS WORKING PAPER' ' DIFFERENT PERSPECTIVES FOR ASSIGNING WEIGHTS TO DETERMINANTS OF HEALTH. (2010). at <<http://www.countyhealthrankings.org/sites/default/files/differentPerspectivesForAssigningWeightsToDeterminantsOfHealth.pdf>>
9. County Health Rankings Model. *County Health Rankings and Roadmaps* at <<https://www.countyhealthrankings.org/explore-health-rankings/measures-data-sources/county-health-rankings-model>>
10. Social Determinants of Health | Healthy People 2020. at <<https://www.healthypeople.gov/2020/topics-objectives/topic/social-determinants-of-health>>
11. CDC Center for Disease Control and prevention. CDC Social Determinants of Health: Know What Affects Health. *cdc.gov* at <<https://www.cdc.gov/socialdeterminants/index.htm>>
12. HealthPeople.gov. *Social Determinants of health* at <<https://www.healthypeople.gov/2020/topics-objectives/topic/social-determinants-of-health>>

13. Braveman, P. & Gruskin, S. Defining equity in health. *J. Epidemiol. Community Health* **57**, 254–258 (2003).
14. Adler, N. E. *et al.* Addressing Social Determinants of Health and Health Disparities. Discussion Paper, Vital Directions for Health and Health Care Series. National Academy of Medicine, Washington, DC. . (2016). at <<https://nam.edu/wp-content/uploads/2016/09/addressingsocial-determinants-of-health-and-health-disparities.pdf>>
15. Orgera, K. & Artiga, S. Disparities in Health and Health Care: Five Key Questions and Answers. at <<http://files.kff.org/attachment/Issue-Brief-Disparities-in-Health-and-Health-Care-Five-Key-Questions-and-Answers>>
16. Minority Health and Health Disparities Research Framework. *National Institute on Minority and Health Disparities. NIMHD* at <<https://www.nimhd.nih.gov/about/overview/research-framework/nimhd-framework.html>>
17. Thornton, R. L. J. *et al.* Evaluating strategies for reducing health disparities by addressing the social determinants of health. *Health Aff. (Millwood)* **35**, 1416–1423 (2016).
18. Gottlieb, L. M. & Alderwick, H. Integrating social and medical care: could it worsen health and increase inequity? *Ann Fam Med* **17**, 77–81 (2019).
19. rr69-545.pdf.
20. A Guide for Planning & Reporting Community Benefit. (2015).
21. Form 990 Schedule H.
22. Young, G. J., Flaherty, S., Zepeda, E. D., Singh, S. R. & Rosen Cramer, G. Community Benefit Spending By Tax-Exempt Hospitals Changed Little After ACA. *Health Aff. (Millwood)* **37**, 121–124 (2018).
23. Association of Academic Health Centers. *About Academic-Health-Centers* at <<http://www.aahcdc.org/About/Academic-Health-Centers>>
24. Brutger, R. Academic Health' ' Centers and Community Health Centers: The Landscape of Current Partnerships. (2010).
25. in *The Health Care Safety Net in a Post-Reform World* (eds. Hall, M. A. & Rosenbaum, S.) 108–125 (Rutgers University Press, 2012). doi:10.36019/9780813553177-007
26. Boelen, C., Heck, J. & World Health Organization. Division of Development of Human Resources for Health. Defining and Measuring' ' the Social Accountability of Medical Schools. (1995).

27. Accountable Health Communities Model. *cms.gov* at <<https://innovation.cms.gov/initiatives/ahcm>>
28. Centers for Medicare and Medicaid Services. Shared savings program. *Shared Savings Program* *www.cms.gov*. at <<https://www-cms-gov.libproxy.lib.unc.edu/Medicare/Medicare-Fee-for-Service-Payment/sharedsavingsprogram/index.html>>
29. CMS.gov. *Centers for Medicare & Medicaid Services Hospital Quality Initiative* at <<https://www.cms.gov/Medicare/Quality-Initiatives-Patient-Assessment-Instruments/HospitalQualityInits/OutcomeMeasures>>
30. Lockner, A. M. & Walcker, C. A. INSIGHT: The Healthcare Industry's Shift from Fee-for-Service to Value-Based Reimbursement. *Bloomberg Law - Health, law and Business news* (2018). at <<https://news-bloomberglaw-com.libproxy.lib.unc.edu/health-law-and-business/insight-the-healthcare-industrys-shift-from-fee-for-service-to-value-based-reimbursement>>
31. Blumenthal, D., McCarthy, D. & Shah, T. B. Academic Medical Centers and High-Need, High-Cost Patients: A Call to Action. *Acad. Med.* **93**, 1617–1619 (2018).
32. The State of Value-based Care in 2018: 10 Key Trends to Know. at <<https://hitconsultant.net/2018/06/18/value-based-care-trends/#.XTyeGpNKi8h>>
33. Bundled Payments for Care Improvement (BPCI) Initiative: General Information | Center for Medicare & Medicaid Innovation. at <<https://innovation.cms.gov/initiatives/bundled-payments/>>
34. Hardin, L., Kilian, A. & Murphy, E. Bundled Payments for Care Improvement: Preparing for the Medical Diagnosis-Related Groups. *J Nurs Adm* **47**, 313–319 (2017).
35. Comprehensive ESRD Care Model | Center for Medicare & Medicaid Innovation. at <<https://innovation.cms.gov/initiatives/comprehensive-esrd-care/>>
36. DiSesa, V. J. & Kaiser, L. R. What's in a name? the necessary transformation of the academic medical center in the era of population health and accountable care. *Acad. Med.* **90**, 842–845 (2015).
37. Halvorson, S. A. C., Tanski, M. E. & Yackel, T. R. Transitioning from volume to value: one academic medical center's approach to improving population health. *Acad. Med.* **92**, 666–670 (2017).
38. Goodman, R. A., Bunnell, R. & Posner, S. F. What is “community health”? Examining the meaning of an evolving field in public health. *Prev. Med.* **67 Suppl 1**, S58–61 (2014).
39. Kindig, D. & Stoddart, G. What is population health? *Am. J. Public Health* **93**, 380–383 (2003).

40. Working Definition of Population Health. *Roundtable on Population Health Improvement The National Academies of Sciences Engineering Medicine* at <<http://nationalacademies.org/hmd/~media/Files/Activity%20Files/PublicHealth/PopulationHealthImprovementRT/Pop%20Health%20RT%20Population%20Health%20Working%20Definition.pdf>>
41. Wilkins, C. H. & Alberti, P. M. Shifting academic health centers from a culture of community service to community engagement and integration. *Acad. Med.* **94**, 763–767 (2019).
42. Wartman, S. A. & Steinberg, M. J. The role of academic health centers in addressing social responsibility. *Med. Teach.* **33**, 638–642 (2011).
43. Perman, J. A., Mullins, C. D. & Newhouse, R. in *The transformation of academic health centers* 195–202 (Elsevier, 2015). doi:10.1016/B978-0-12-800762-4.00019-0
44. Knettel, A. The Business Case' ' for Academic Health Centers Addressing Environmental, Social, and Behavioral Determinants of Health. (2011).
45. Schroeder, S. A. Social Justice as the Moral Core of Family Medicine: A Perspective from the Keystone IV Conference. *J Am Board Fam Med* **29 Suppl 1**, S69–71 (2016).
46. Alberti, P. M. *et al.* Communities, social justice, and academic health centers. *Acad. Med.* **93**, 20–24 (2018).
47. Wu, A. W. *et al.* The Baltimore Community-Based Organizations Neighborhood Network: Enhancing Capacity Together (CONNECT) Cluster RCT. *Am. J. Prev. Med.* **57**, e31–e41 (2019).
48. Navathe, A. S. *et al.* Hospital Readmission and Social Risk Factors Identified from Physician Notes. *Health Serv. Res.* **53**, 1110–1136 (2018).
49. Graham, K. L., Wilker, E. H., Howell, M. D., Davis, R. B. & Marcantonio, E. R. Differences between early and late readmissions among patients: a cohort study. *Ann. Intern. Med.* **162**, 741–749 (2015).
50. Hatef, E. *et al.* Assessing the Impact of Social Needs and Social Determinants of Health on Health Care Utilization: Using Patient- and Community-Level Data. *Popul. Health Manag.* (2020). doi:10.1089/pop.2020.0043
51. Lin, M. P. *et al.* ED-based care coordination reduces costs for frequent ED users. *Am. J. Manag. Care* **23**, 762–766 (2017).
52. Zhang, Y. *et al.* Assessing the impact of social determinants of health on predictive models for potentially avoidable 30-day readmission or death. *PLoS One* **15**, e0235064 (2020).

53. Rubin, D. J., Donnell-Jackson, K., Jhingan, R., Golden, S. H. & Paranjape, A. Early readmission among patients with diabetes: a qualitative assessment of contributing factors. *J Diabetes Complicat* **28**, 869–873 (2014).
54. Georgiev, R., Stryckman, B. & Velez, R. The integral role of nurse practitioners in community paramedicine. *The Journal for Nurse Practitioners* **15**, 725–731 (2019).
55. Gonzalo, J. D., Chang, A. & Wolpaw, D. R. New educator roles for health systems science: implications of new physician competencies for U.S. medical school faculty. *Acad. Med.* **94**, 501–506 (2019).
56. Bedoya, P. *et al.* Student hotspotting: teaching the interprofessional care of complex patients. *Acad. Med.* **93**, 56–59 (2018).
57. Aguilar-Gaxiola, S. *et al.* Towards a unified taxonomy of health indicators: academic health centers and communities working together to improve population health. *Acad. Med.* **89**, 564–572 (2014).
58. DeVoe, J. E., Likumahuwa-Ackman, S., Shannon, J. & Steiner Hayward, E. Creating 21st-Century Laboratories and Classrooms for Improving Population Health: A Call to Action for Academic Medical Centers. *Acad. Med.* **92**, 475–482 (2017).
59. Michener, L. *et al.* Aligning the goals of community-engaged research: why and how academic health centers can successfully engage with communities to improve health. *Acad. Med.* **87**, 285–291 (2012).
60. Rieselbach, R. E. *et al.* A new community health center/academic medicine partnership for medicaid cost control, powered by the mega teaching health center. *Acad. Med.* **93**, 406–413 (2018).
61. National Center for Advancing Translational Science. Clinical and Translational Science Awards (CTSA) Program. *National Center for Advancing Translational Science* at <<https://ncats.nih.gov/ctsa>>
62. Berkowitz, S. A. *et al.* Case Study: Johns Hopkins Community Health Partnership: A model for transformation. *Healthc (Amst)* **4**, 264–270 (2016).
63. Oostra, R. Strengthening community through strategy: building a premier academic medical destination. *J. Healthc. Manag.* **63**, 90–93 (2018).
64. Fagnan, L. J. Moving Upstream-Health Extension and Primary Care. *J Am Board Fam Med* **30**, 10–12 (2017).
65. Battaglia, T. A. *et al.* Translating research into practice: Protocol for a community-engaged, stepped wedge randomized trial to reduce disparities in breast cancer treatment through a regional patient navigation collaborative. *Contemp. Clin. Trials* **93**, 106007 (2020).

66. Kaplan, S. A. & Gourevitch, M. N. Leveraging population health expertise to enhance community benefit. *Front Public Health* **8**, 88 (2020).
67. Tung, E. L. *et al.* Experiences of Community Violence Among Adults with Chronic Conditions: Qualitative Findings from Chicago. *J. Gen. Intern. Med.* **33**, 1913–1920 (2018).
68. Wynn, N., Staffileno, B. A., Grenier, J. M. & Phillips, J. Implementing a food is medicine program to address food insecurity in an academic medical center. *J. Nurs. Care Qual.* **36**, 262–268 (2021).
69. NowPow. at <<https://www.nowpow.com/>>
70. Meyer, D., Lerner, E., Phillips, A. & Zumwalt, K. Universal screening of social determinants of health at a large US academic medical center, 2018. *Am. J. Public Health* **110**, S219–S221 (2020).
71. Fiori, K. *et al.* From policy statement to practice: integrating social needs screening and referral assistance with community health workers in an urban academic health center. *J. Prim. Care Community Health* **10**, 2150132719899207 (2019).
72. Denham, A. C., Hay, S. S., Steiner, B. D. & Newton, W. P. Academic health centers and community health centers partnering to build a system of care for vulnerable patients: lessons from Carolina Health Net. *Acad. Med.* **88**, 638–643 (2013).
73. Rosenbaum, S., Kindig, D. A., Bao, J., Byrnes, M. K. & O’Laughlin, C. The value of the nonprofit hospital tax exemption was \$24.6 billion in 2011. *Health Aff. (Millwood)* **34**, 1225–1233 (2015).
74. Young, G. J., Chou, C.-H., Alexander, J., Lee, S.-Y. D. & Raver, E. Provision of community benefits by tax-exempt U.S. hospitals. *N. Engl. J. Med.* **368**, 1519–1527 (2013).
75. Leider, J. P. *et al.* Establishing a Baseline: Community Benefit Spending by Not-for-Profit Hospitals Prior to Implementation of the Affordable Care Act. *J. Public Health Manag. Pract.* **23**, e1–e9 (2017).
76. Horwitz, L. I., Chang, C., Arcilla, H. N. & Knickman, J. R. Quantifying Health Systems’ Investment In Social Determinants Of Health, By Sector, 2017-19. *Health Aff. (Millwood)* **39**, 192–198 (2020).
77. Fos, E. B., Thompson, M. E., Elnitsky, C. A. & Platonova, E. A. Community Benefit Spending Among North Carolina’s Tax-Exempt Hospitals After Performing Community Health Needs Assessments. *J. Public Health Manag. Pract.* **25**, E1–E8 (2019).
78. Gottlieb, L. M., Wing, H. & Adler, N. E. A systematic review of interventions on patients’ social and economic needs. *Am. J. Prev. Med.* **53**, 719–729 (2017).

79. NC Department of Health and Human Services. About Healthy Opportunities. at <<https://www.ncdhhs.gov/about/departments-initiatives/healthy-opportunities/about-healthy-opportunities>>
80. United health Foundation. America's Health Rankings - NC Summary 2019. at <<https://www.americashealthrankings.org/explore/annual/state/NC>>
81. What Expanding Medicaid Would Mean for North Carolina. *NCDHHS Blog* (2022). at <<https://www.ncdhhs.gov/blog/2022/06/09/what-expanding-medicaid-would-mean-north-carolina>>
82. North Carolina Healthcare Association. North Carolina Healthcare Association - Community Benefit. at <<https://www.ncha.org/community-benefit/>>
83. North Carolina Healthcare Association (NCHA) Community Benefit. at <<https://www.ncha.org/community-benefit/>>
84. GuideStar by Candid - Connecting you with the nonprofit information you need. at <<https://www.guidestar.org/>>
85. Academic medical center teaching hospitals.
86. Internal Revenue Service. Instructions for Schedule H (Form 990). at <<https://www.irs.gov/pub/irs-pdf/i990sh.pdf>>
87. To Improve Health Outside of Hospital Walls, Atrium Health Commits \$10 Million to Affordable Housing. *Atrium Health news* (2019). at <<https://atriumhealth.org/about-us/newsroom/news/2019/06/affordable-housing-commitment>>
88. North Carolina Department of Health and Human Services. NCCARE360. *NCDHHS* at <<https://www.ncdhhs.gov/about/departments-initiatives/healthy-opportunities/nccare360>>
89. The Joint Commission. New Requirements to Reduce Health Care Disparities. *R3 Report Requirement, Rationale, Reference* (2022).
90. CMS Framework for Health Equity_2022 04 06_0.pdf. (2022).
91. NCDHHS. North Carolina's Transformation to Medicaid Managed Care. at <<https://medicaid.ncdhhs.gov/transformation>>
92. North Carolina Department of Health and Human Services. Mission, Vision. *NCDHHS* at <<https://www.ncdhhs.gov/about/dhhs-mission-vision-values-and-goals/>>
93. Status of State Medicaid Expansion Decisions: Interactive Map. *KFF Kaiser Family Foundation* (2022). at <<https://www.kff.org/medicaid/issue-brief/status-of-state-medicaid-expansion-decisions-interactive-map/>>

94. NC FIT Program. *UNC School of medicine Family Medicine* at <https://www.med.unc.edu/fammed/service-to-the-community/clinical-care/formerly-incarcerated-transition-program/>
95. Nundy, S., Cooper, L. A. & Mate, K. S. The quintuple aim for health care improvement: A new imperative to advance health equity. *JAMA* **327**, 521–522 (2022).
96. MacDonald, MD, J. *et al.* Health Technology for All: An Equity-Based Paradigm Shift Opportunity. *NAM Perspectives* (2022).
97. The 8-Step Process for Leading Change - Kotter. at <https://www.kotterinc.com/8-steps-process-for-leading-change/>
98. National Academies of Sciences, Engineering, and Medicine; Health and Medicine Division; Board on Health Care Services; Committee on Integrating Social Needs Care into the Delivery of Health Care to Improve the Nation's Health. *Integrating Social Care into the Delivery of Health Care: Moving Upstream to Improve the Nation's Health*. (National Academies Press (US), 2019). doi:10.17226/25467

ProQuest Number: 30314845

INFORMATION TO ALL USERS

The quality and completeness of this reproduction is dependent on the quality and completeness of the copy made available to ProQuest.



Distributed by ProQuest LLC (2023).

Copyright of the Dissertation is held by the Author unless otherwise noted.

This work may be used in accordance with the terms of the Creative Commons license or other rights statement, as indicated in the copyright statement or in the metadata associated with this work. Unless otherwise specified in the copyright statement or the metadata, all rights are reserved by the copyright holder.

This work is protected against unauthorized copying under Title 17,
United States Code and other applicable copyright laws.

Microform Edition where available © ProQuest LLC. No reproduction or digitization of the Microform Edition is authorized without permission of ProQuest LLC.

ProQuest LLC
789 East Eisenhower Parkway
P.O. Box 1346
Ann Arbor, MI 48106 - 1346 USA